VIT BHOPAL UNIVERSITY

School of Computing Science and Engineering

Bhopal-Indore Highway, Kothrikalan, Sehore

Madhya Pradesh - 466114

CSE3016 – AWS SOLUTION ARCHITECT

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BRANCH : AIML

SEMESTER: INTERIM SEMESTER 2025-26

INDEX

EXP.NO: 01	Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware
DATE: 10-11-25	

AIM

To Launch an EC2 instance using the AWS Management Console and to successfully deploy a website using Apache middle ware

PROCEDURE

1. Launch EC2 Instance: Use the AWS Console to launch a t2.micro instance with Amazon Linux 2023 and an appropriate Key Pair.

2. Configure Security Group: Ensure the instance's Security Group allows inbound traffic on Port 22 (SSH) and Port 80 (HTTP).

3. Connect via SSH: Use your downloaded .pem key file to SSH into the instance using the public IP address.

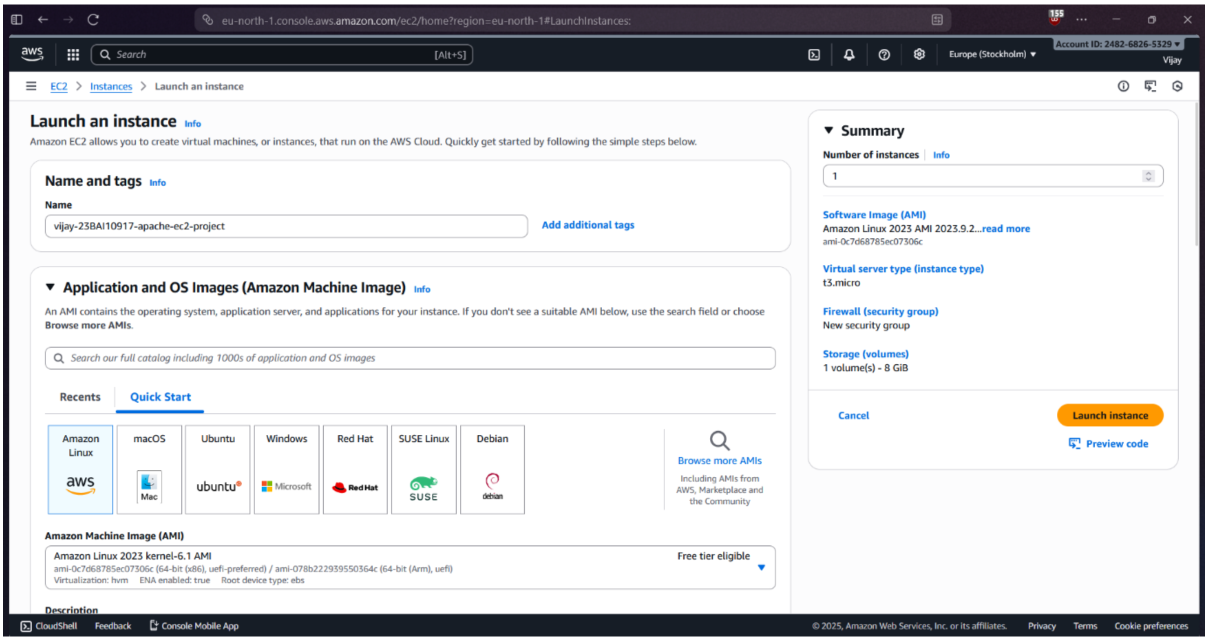
4. Install Apache: Run sudo dnf update -y then sudo dnf install httpd -y on the instance.

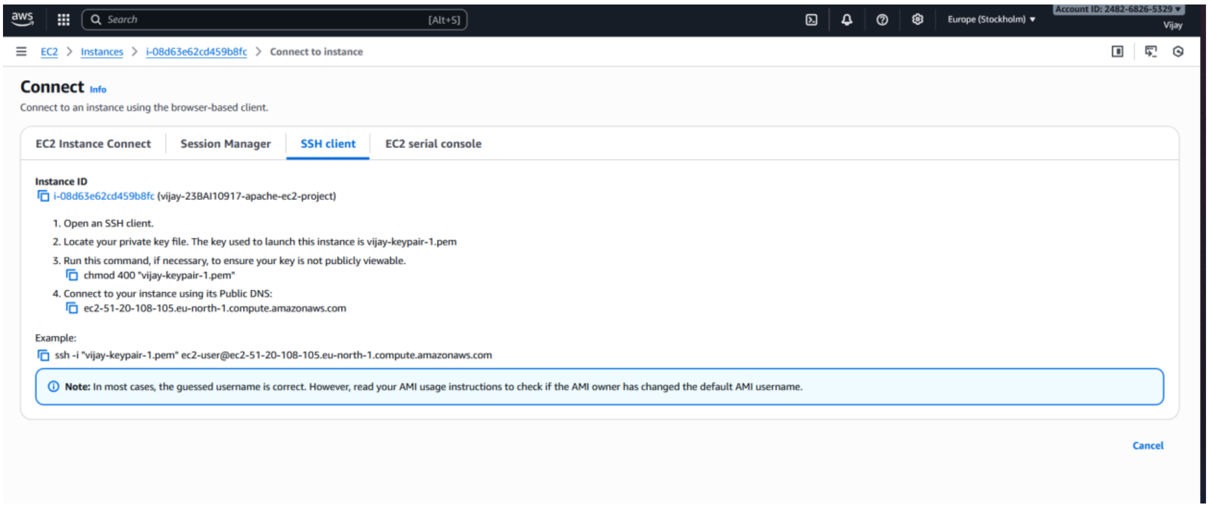
5. Start and Enable Service: Run sudo systemctl start httpd and sudo systemctl enable httpd.

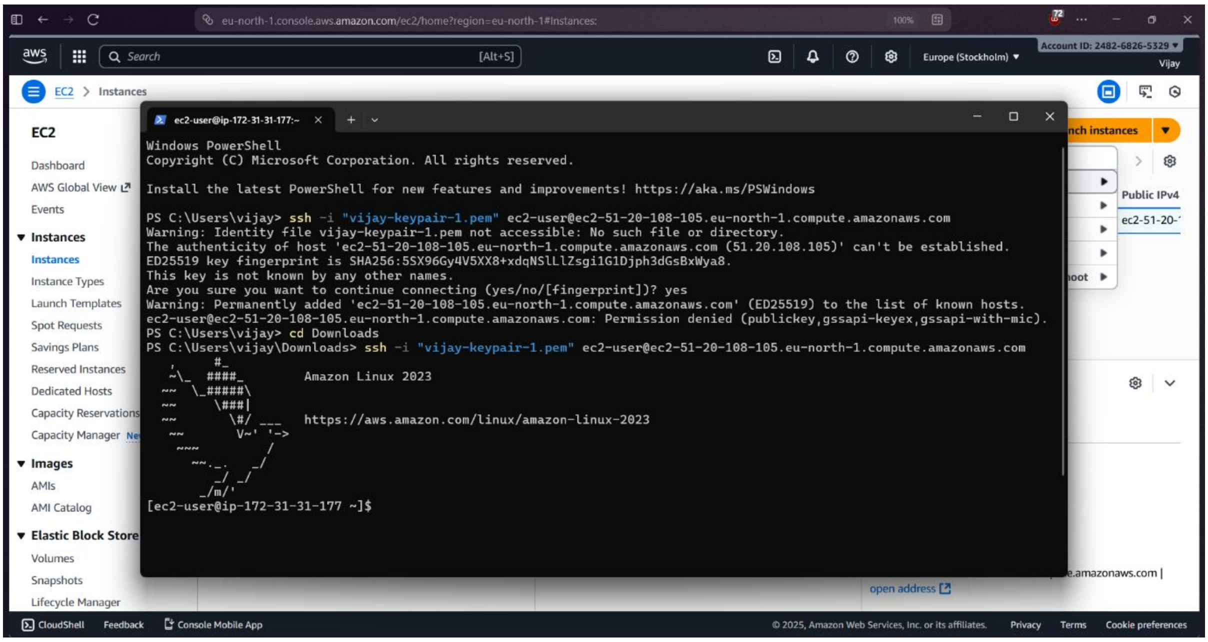
6. Deploy Website: Navigate to /var/www/html/ and create or upload your website files (e.g., sudo nano index.html).

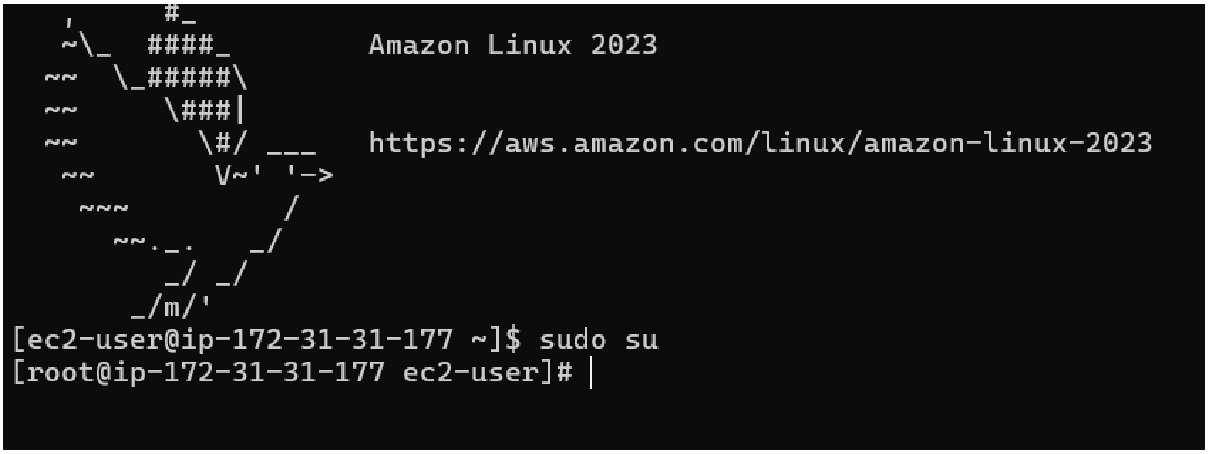
7. Test: Open the instance's Public IPv4 Address in a web browser to view the deployed website.

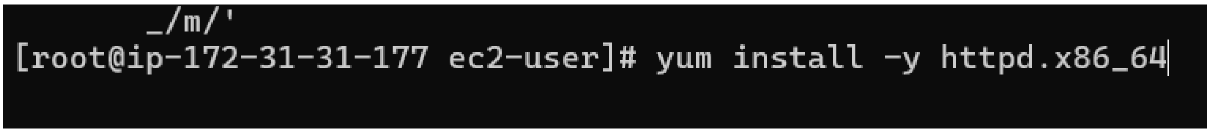
IMPLEMENTATION SCREENSHOT





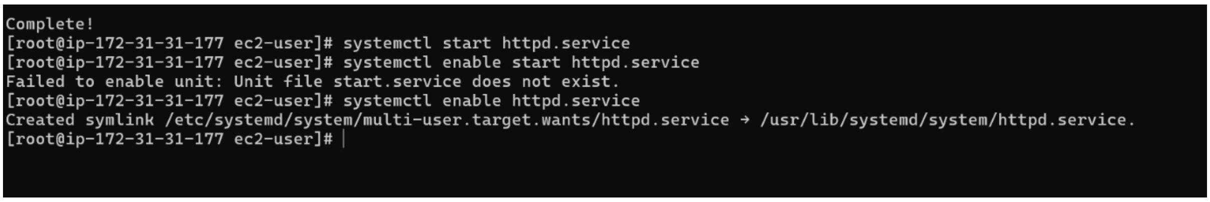

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:
Search
[Alt+S]
Europe (Stockholm)
Account ID: 2482-6826-5329
Vijay
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Public IPv4
open address
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
ec2-user@ip-172-31-31-177:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\vijay> ssh -i "vijay-keypair-1.pem" ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com
Warning: Identity file vijay-keypair-1.pem not accessible: No such file or directory.
The authenticity of host 'ec2-51-20-108-105.eu-north-1.compute.amazonaws.com (51.20.108.105)' can't be established.
ED25519 key fingerprint is SHA256:5SX96Gy4V5XX8+xdqNSlLlZsgi1G1Djph3dGsBxWya8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-51-20-108-105.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
PS C:\Users\vijay> cd Downloads
PS C:\Users\vijay\Downloads> ssh -i "vijay-keypair-1.pem" ec2-user@ec2-51-20-108-105.eu-north-1.compute.amazonaws.com
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-31-177 ~]$

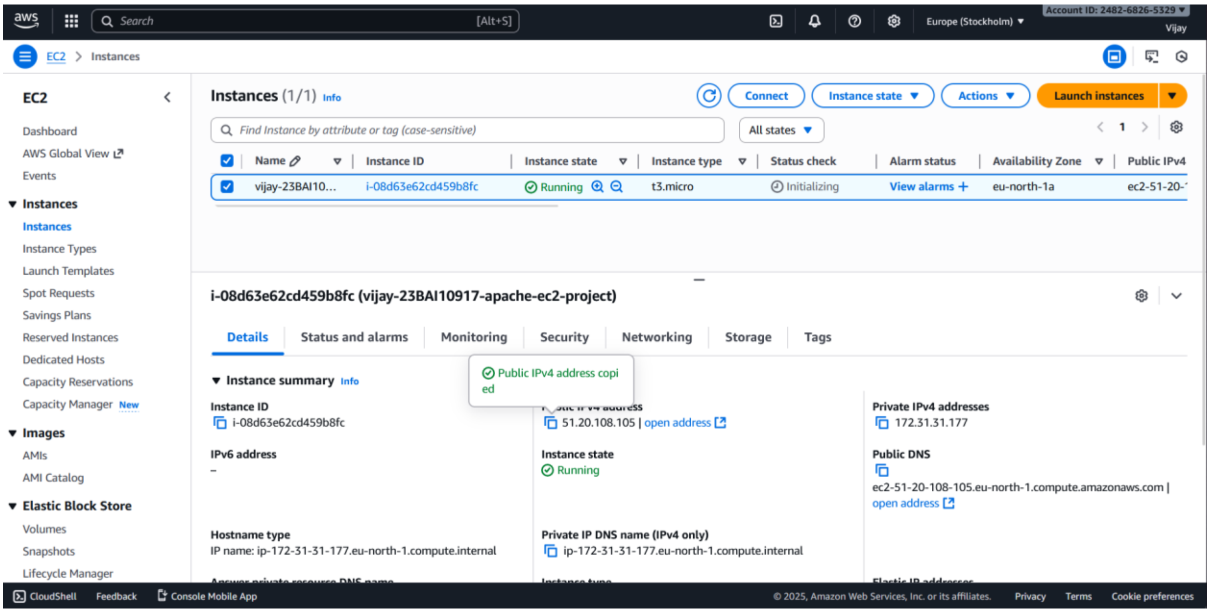

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-31-177 ~]$ sudo su
[root@ip-172-31-31-177 ec2-user]#

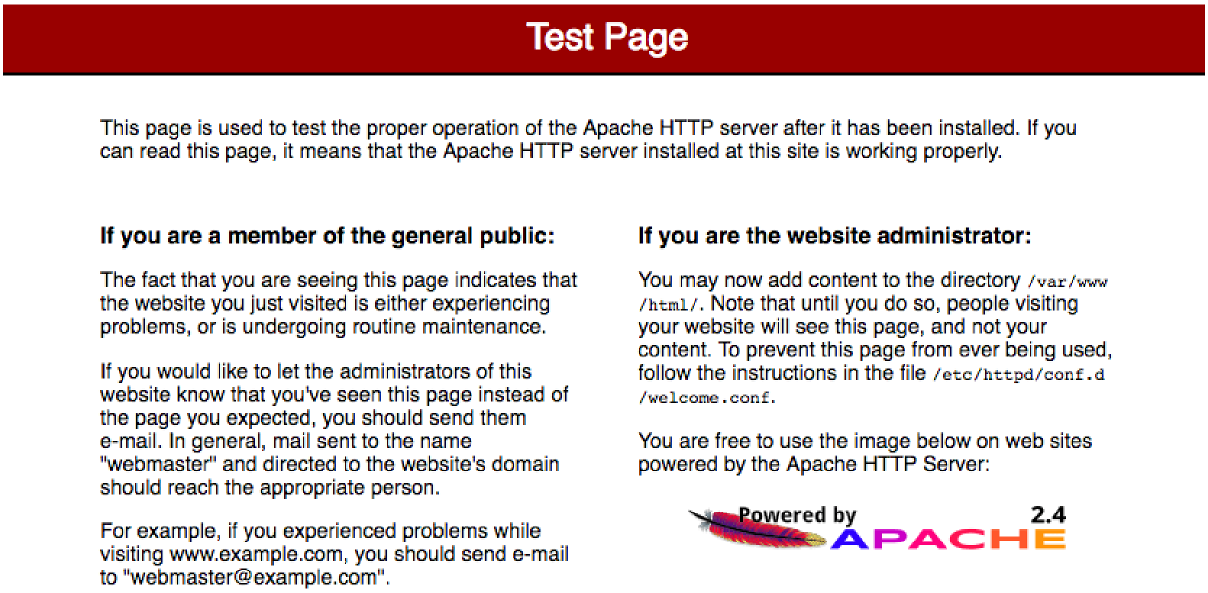

[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64

```
[root@ip-172-31-31-177 ec2-user]# yum install -y httpd.x86_64
Amazon Linux 2023 Kernel Livepatch repo 273 kB/s |  29 kB     00:00
Dependencies resolved.
=======================================================================
 Package                Arch   Version                   Repository   Size
=======================================================================
Installing:
 httpd                  x86_64 2.4.65-1.amzn2023.0.2     amazonlinux  47 k
Installing dependencies:
 apr                    x86_64 1.7.5-1.amzn2023.0.4      amazonlinux 129 k
 apr-util               x86_64 1.6.3-1.amzn2023.0.2      amazonlinux  97 k
 apr-util-lmdb          x86_64 1.6.3-1.amzn2023.0.2      amazonlinux  13 k
 generic-logos-httpd    noarch 18.0.0-12.amzn2023.0.3    amazonlinux  19 k
 httpd-core             x86_64 2.4.65-1.amzn2023.0.2     amazonlinux 1.4 M
 httpd-filesystem       noarch 2.4.65-1.amzn2023.0.2     amazonlinux  13 k
 httpd-tools            x86_64 2.4.65-1.amzn2023.0.2     amazonlinux  81 k
 libbrotli              x86_64 1.0.9-4.amzn2023.0.2      amazonlinux 315 k
 mailcap                noarch 2.1.49-3.amzn2023.0.3     amazonlinux  33 k
Installing weak dependencies:
 apr-util-openssl       x86_64 1.6.3-1.amzn2023.0.2      amazonlinux  15 k
 mod_http2              x86_64 2.0.27-1.amzn2023.0.3     amazonlinux 166 k
 mod_lua                x86_64 2.4.65-1.amzn2023.0.2     amazonlinux  60 k

Transaction Summary
=======================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-lmdb-1.6.3-1.amzn2023. 369 kB/s |  13 kB     00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64 3.0 MB/s | 129 kB     00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x 2.1 MB/s |  97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn20 737 kB/s |  15 kB     00:00
(5/13): httpd-2.4.65-1.amzn2023.0.2.x86 2.5 MB/s |  47 kB     00:00
(6/13): generic-logos-httpd-18.0.0-12.a 786 kB/s |  19 kB     00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2 578 kB/s |  13 kB     00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0 2.9 MB/s |  81 kB     00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0.  28 MB/s | 1.4 MB     00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2 9.9 MB/s | 315 kB     00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3. 1.3 MB/s |  33 kB     00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0. 6.4 MB/s | 166 kB     00:00
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s |  60 kB     00:00
-----------------------------------------------------------------------
Total                                    14 MB/s | 2.4 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64               1/13
  Installing       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64     2/13
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_    3/13
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64          4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch          5/13
```

```
(13/13): mod_lua-2.4.65-1.amzn2023.0.2. 2.8 MB/s |  60 kB     00:00
--------------------------------------------------------------------------------
Total                                           14 MB/s | 2.4 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                       1/13
  Installing       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64             2/13
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_            3/13
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64                  4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                  5/13
  Installing       : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64              6/13
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                 7/13
  Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Installing       : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Installing       : httpd-core-2.4.65-1.amzn2023.0.2.x86_64               9/13
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64               10/13
  Installing       : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                 11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3           12/13
  Installing       : httpd-2.4.65-1.amzn2023.0.2.x86_64                   13/13
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64                   13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                       1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                  2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64             3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_            4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3            5/13
  Verifying        : httpd-2.4.65-1.amzn2023.0.2.x86_64                    6/13
  Verifying        : httpd-core-2.4.65-1.amzn2023.0.2.x86_64               7/13
  Verifying        : httpd-filesystem-2.4.65-1.amzn2023.0.2.noa            8/13
  Verifying        : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64              9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                 11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64               12/13
  Verifying        : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                 13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-2.4.65-1.amzn2023.0.2.x86_64
  httpd-core-2.4.65-1.amzn2023.0.2.x86_64
  httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
  httpd-tools-2.4.65-1.amzn2023.0.2.x86_64
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-31-177 ec2-user]#
```


Complete!
[root@ip-172-31-31-177 ec2-user]# systemctl start httpd.service
[root@ip-172-31-31-177 ec2-user]# systemctl enable start httpd.service
Failed to enable unit: Unit file start.service does not exist.
[root@ip-172-31-31-177 ec2-user]# systemctl enable httpd.service
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-31-177 ec2-user]#


Search
[Alt+S]
Europe (Stockholm)
Account ID: 2482-6826-5329
Vijay
EC2
Instances
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
vijay-23BAI10...
i-08d63e62cd459b8fc
Running
t3.micro
Initializing
View alarms +
eu-north-1a
i-08d63e62cd459b8fc (vijay-23BAI10917-apache-ec2-project)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Public IPv4 address copied
Instance ID
i-08d63e62cd459b8fc
51.20.108.105 | open address
Private IPv4 addresses
172.31.31.177
IPv6 address
–
Instance state
Running
Public DNS
ec2-51-20-108-105.eu-north-1.compute.amazonaws.com |
open address
Hostname type
IP name: ip-172-31-31-177.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-31-177.eu-north-1.compute.internal
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the Apache HTTP server installed at this site is working properly.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting www.example.com, you should send e-mail to "webmaster@example.com".

If you are the website administrator:

You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

You are free to use the image below on web sites powered by the Apache HTTP Server:

Powered by APACHE 2.4

RESULT

Thus, EC2 instance has been launched using the AWS Management Console and a website is deployed using the Apache Middle ware

EXP.NO: 02	Deploy a static website in S3 and manage the website using S3 life cycle management
DATE: 11-11-25	

AIM

To deploy a static website in S3 and manage the website using S3 Life Cycle Management, in the AWS Management Console

PROCEDURE

Step 1: Create and Configure the S3 Bucket

- Create Bucket: In the AWS S3 Console, click "Create bucket". Give it a globally unique name (e.g., my-static-site).

- Public Access: Uncheck "Block all public access" during creation (you'll set permissions next).

- Enable Website Hosting: Go to the bucket's Properties tab, find Static website hosting, click "Edit", select "Enable", and specify your Index document (e.g., index.html) and Error document (e.g., error.html). Save changes.

Step 2: Upload Files and Set Permissions

- Upload Content: Go to the Objects tab and Upload all your website files (HTML, CSS, images).

- Apply Bucket Policy: Go to the Permissions tab and click "Edit" under Bucket Policy. Paste a policy granting public s3:GetObject access to all resources in your bucket, replacing YOUR_BUCKET_NAME. Save changes.

- Test Access: Paste the Bucket website endpoint (found under Properties) into a browser to confirm the site is live.

- Step 3: Implement Lifecycle Management
- Create Rule: Go to the Management tab, find Lifecycle rules, and click "Create lifecycle rule".

- Define Scope: Give the rule a name (e.g., Archive-and-Expire) and choose "Apply to all objects" or use a prefix (e.g., backups/) to target specific folders.

- Configure Transitions (Cost Optimization): Check "Transition current versions of objects...".

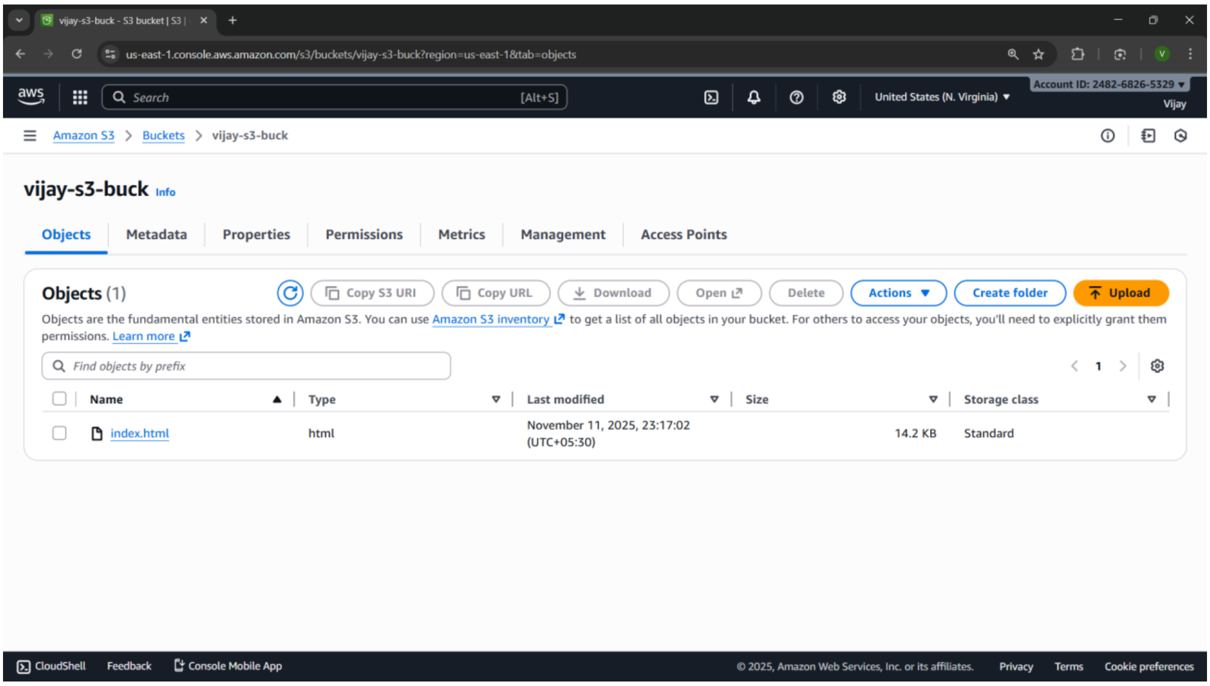
- Click "Add transition": Move objects to Standard-IA after 30 days.

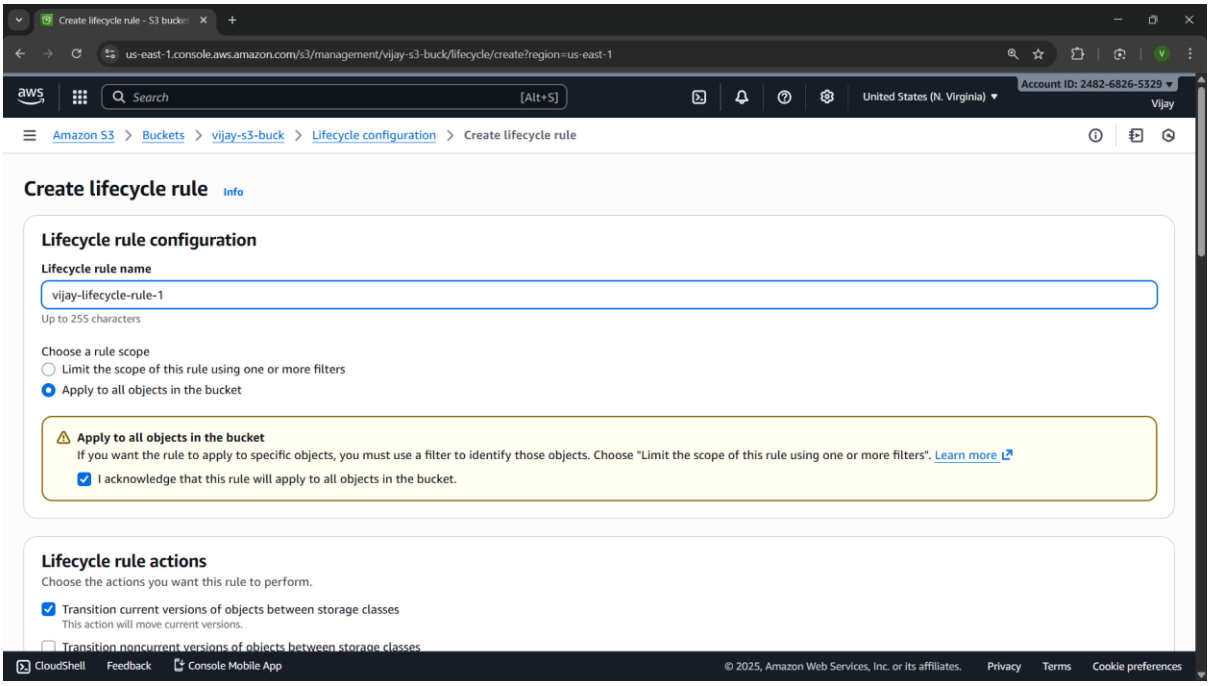
- Click "Add transition": Move objects to Glacier Instant Retrieval after 90 days.

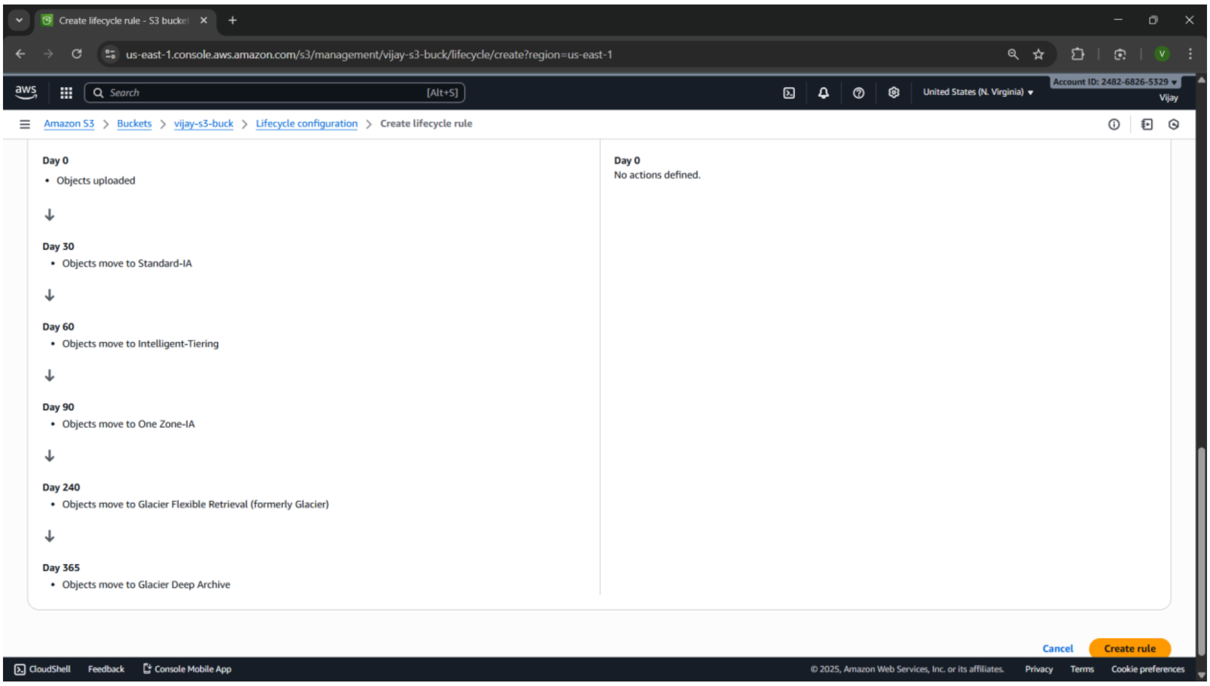
- Configure Expiration (Cleanup): Check "Expire current versions of objects" and set the number of days after creation (e.g., 365 days) to permanently delete the objects.

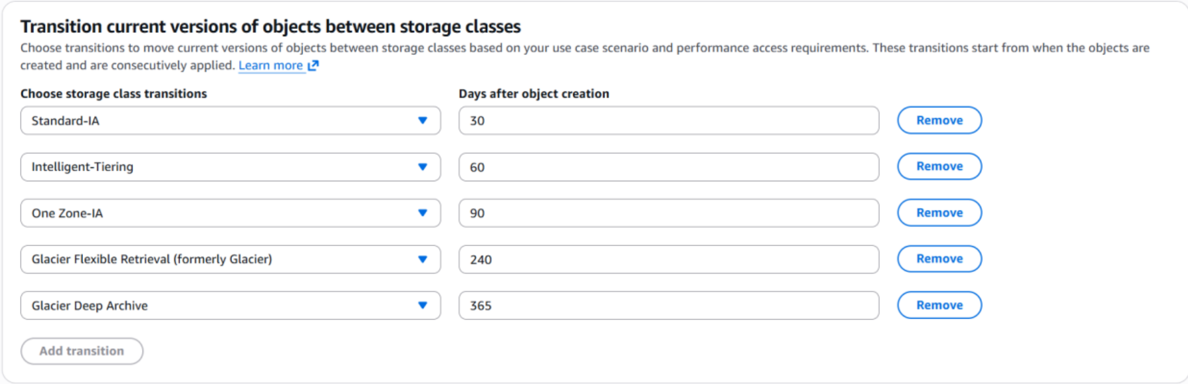
- Finalize: Click "Create rule". The lifecycle rule will now automatically manage the storage class and eventual deletion of your objects based on their age.

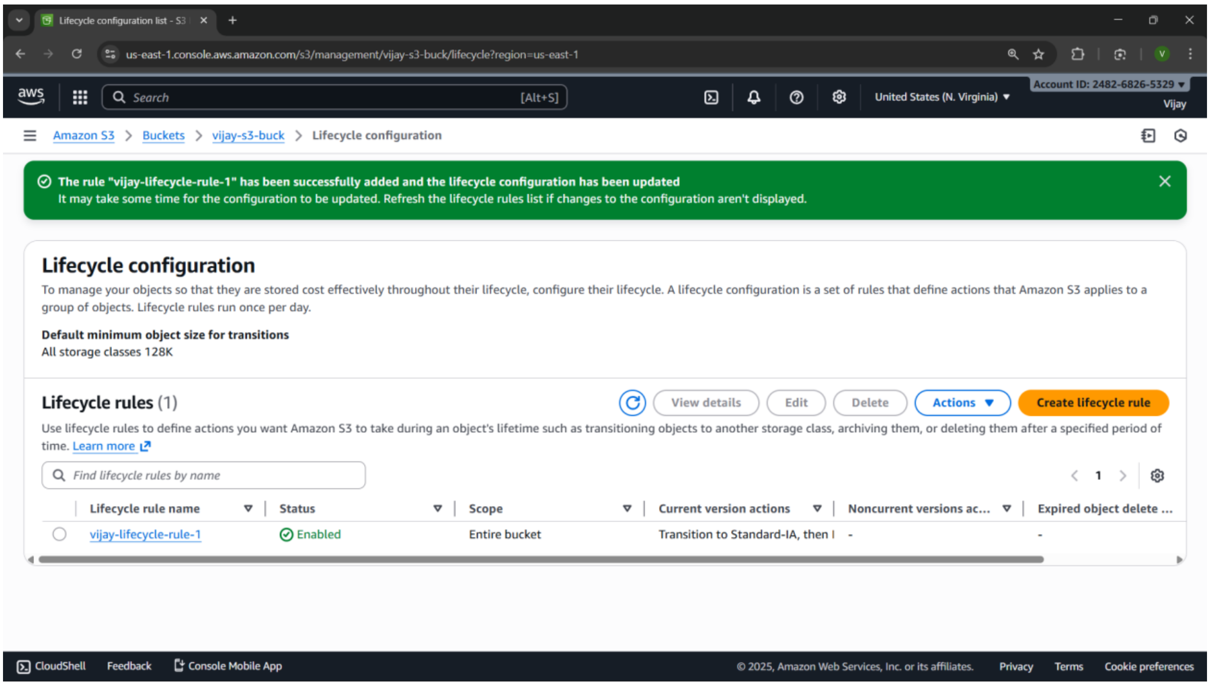
IMPLEMENTATION SCREENSHOT




Create lifecycle rule - S3 bucket
us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule
Create lifecycle rule Info
Lifecycle rule configuration
Lifecycle rule name
vijay-lifecycle-rule-1
Up to 255 characters
Choose a rule scope
Limit the scope of this rule using one or more filters
Apply to all objects in the bucket
Apply to all objects in the bucket
If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". Learn more
I acknowledge that this rule will apply to all objects in the bucket.
Lifecycle rule actions
Choose the actions you want this rule to perform.
Transition current versions of objects between storage classes
This action will move current versions.
Transition noncurrent versions of objects between storage classes
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Create lifecycle rule - S3 bucket
us-east-1.console.aws.amazon.com/s3/management/vijay-s3-buck/lifecycle/create?region=us-east-1
Search
[Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
Amazon S3 > Buckets > vijay-s3-buck > Lifecycle configuration > Create lifecycle rule
Day 0
Objects uploaded
Day 30
Objects move to Standard-IA
Day 60
Objects move to Intelligent-Tiering
Day 90
Objects move to One Zone-IA
Day 240
Objects move to Glacier Flexible Retrieval (formerly Glacier)
Day 365
Objects move to Glacier Deep Archive
Day 0
No actions defined.
Cancel
Create rule
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences





RESULT

Thus, a static website is deployed in S3 and it is managed using S3 life cycle management.

EXP.NO: 03	Creating DNS records, configuring routing policies, implementing health checks, and monitoring DNS health using Route 53
DATE: 12-11-25	

AIM

To establish a low-latency, and highly available Domain Name System (DNS) architecture for a web application using AWS Route 53, including the configuration of various routing policies, implementation of active health checks for continuous resource monitoring, and integration with CloudWatch for automated DNS health monitoring and alerting capabilities.

PROCEDURE

1. Create Hosted Zone: In Route 53, create a Public Hosted Zone for your domain (example.com).

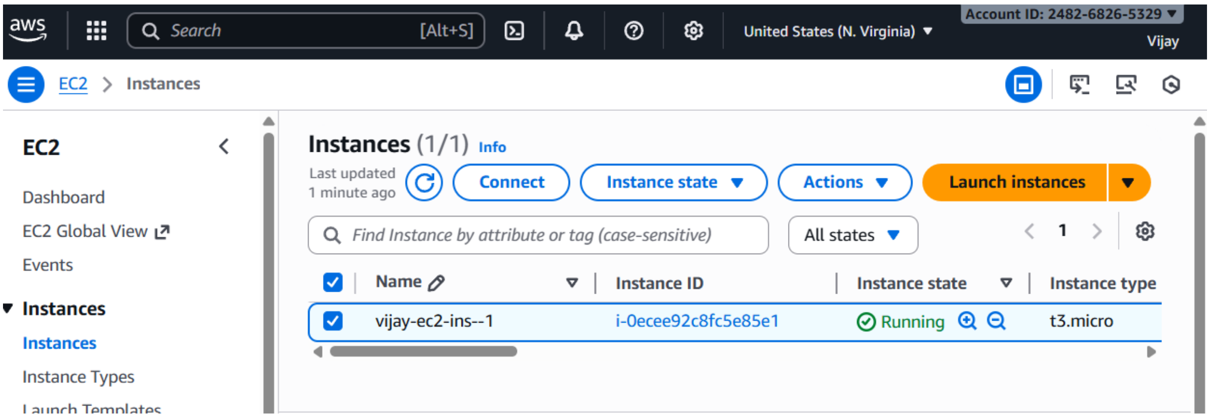
2. Create Resource Records: Create an A record (or CNAME/ALIAS) pointing to your resource (e.g., EC2/Load Balancer IP). Set the Routing Policy (e.g., Simple or Failover).

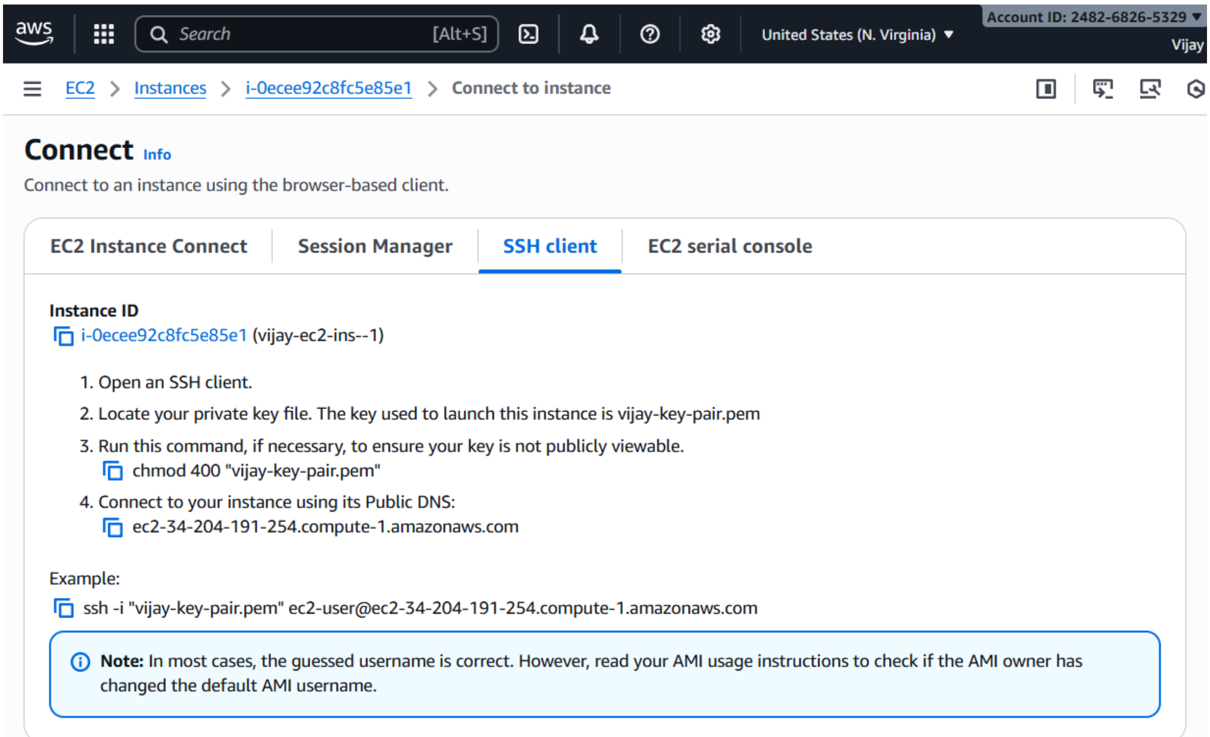
3. Implement Health Check: Create a Health Check to monitor your endpoint (IP address, domain, port, and protocol, e.g., HTTP on port 80).

4. Configure Routing: For Failover or Weighted routing, edit the resource record and associate it with the new Health Check.

5. Set Monitoring/Alerts: Select the Health Check and create a CloudWatch Alarm (e.g., using the HealthCheckStatus metric) to send an SNS notification if the status becomes Unhealthy.

IMPLEMENTATION SCREENSHOT





```
ec2-user@ip-172-31-72-108:~

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\vijay> ssh -i "vijay-key-pair.pem" ec2-user@ec2-34-204-191-254.compute-1.amazonaws.com
Warning: Identity file vijay-key-pair.pem not accessible: No such file or directory.
The authenticity of host 'ec2-34-204-191-254.compute-1.amazonaws.com (34.204.191.254)' can't be establ
ished.
ED25519 key fingerprint is SHA256:yjZYJPzlmcSWmKzw9KJQsUZi2H/K9hAeWAWOKfsO6KA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-204-191-254.compute-1.amazonaws.com' (ED25519) to the list of known
 hosts.
ec2-user@ec2-34-204-191-254.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-
with-mic).
PS C:\Users\vijay> cd Downloads
PS C:\Users\vijay\Downloads> ssh -i "vijay-key-pair.pem" ec2-user@ec2-34-204-191-254.compute-1.amazona
ws.com
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-72-108 ~]$
```
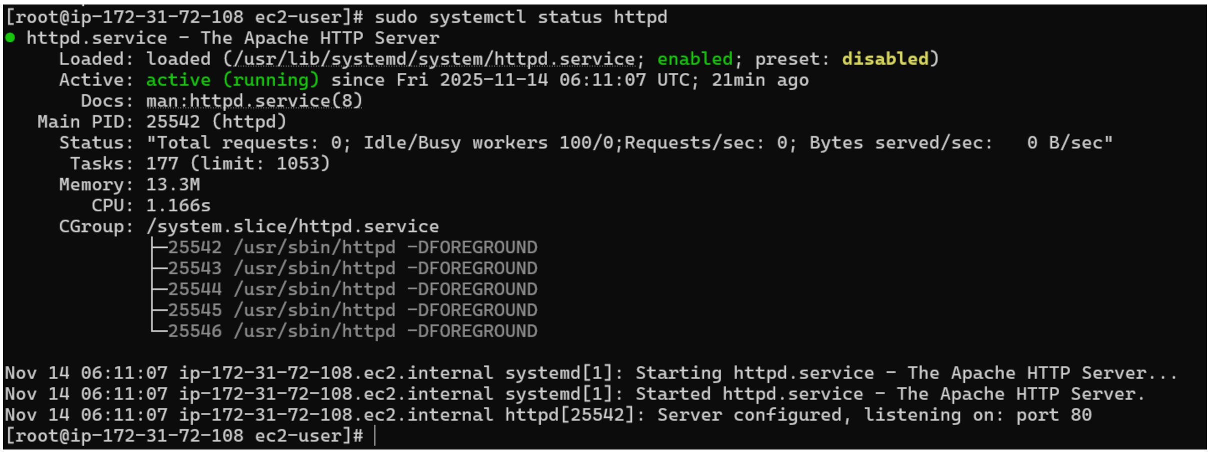

```
[ec2-user@ip-172-31-72-108 ~]$ sudo su
```

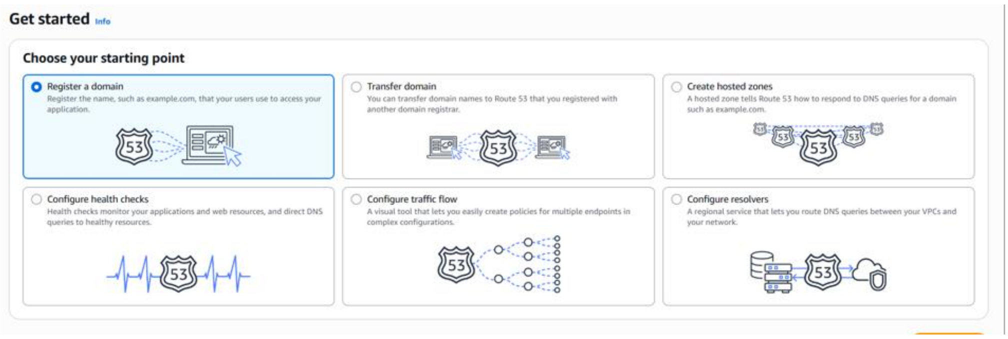
```
[root@ip-172-31-72-108 ec2-user]# yum install httpd -y
Amazon Linux 2023 Kernel Livepatch repository                                    278 kB/s |  29 kB     00:00
Dependencies resolved.
===========================================================================================================
 Package                          Architecture       Version                            Repository          Size
===========================================================================================================
Installing:
 httpd                            x86_64             2.4.65-1.amzn2023.0.2              amazonlinux         47 k
Installing dependencies:
 apr                              x86_64             1.7.5-1.amzn2023.0.4               amazonlinux        129 k
 apr-util                         x86_64             1.6.3-1.amzn2023.0.2               amazonlinux         97 k
 apr-util-lmdb                    x86_64             1.6.3-1.amzn2023.0.2               amazonlinux         13 k
 generic-logos-httpd              noarch             18.0.0-12.amzn2023.0.3             amazonlinux         19 k
 httpd-core                       x86_64             2.4.65-1.amzn2023.0.2              amazonlinux        1.4 M
 httpd-filesystem                 noarch             2.4.65-1.amzn2023.0.2              amazonlinux         13 k
 httpd-tools                      x86_64             2.4.65-1.amzn2023.0.2              amazonlinux         81 k
 libbrotli                        x86_64             1.0.9-4.amzn2023.0.2               amazonlinux        315 k
 mailcap                          noarch             2.1.49-3.amzn2023.0.3              amazonlinux         33 k
Installing weak dependencies:
 apr-util-openssl                 x86_64             1.6.3-1.amzn2023.0.2               amazonlinux         15 k
 mod_http2                        x86_64             2.0.27-1.amzn2023.0.3              amazonlinux        166 k
 mod_lua                          x86_64             2.4.65-1.amzn2023.0.2              amazonlinux         60 k

Transaction Summary
===========================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                                 2.7 MB/s |  97 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm                            346 kB/s |  13 kB     00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                      2.9 MB/s | 129 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm                         734 kB/s |  15 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                    822 kB/s |  19 kB     00:00
(6/13): httpd-2.4.65-1.amzn2023.0.2.x86_64.rpm                                   2.1 MB/s |  47 kB     00:00
(7/13): httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch.rpm                        604 kB/s |  13 kB     00:00
(8/13): httpd-tools-2.4.65-1.amzn2023.0.2.x86_64.rpm                             3.5 MB/s |  81 kB     00:00
(9/13): httpd-core-2.4.65-1.amzn2023.0.2.x86_64.rpm                               33 MB/s | 1.4 MB     00:00
(10/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                                1.5 MB/s |  33 kB     00:00
(11/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                               8.5 MB/s | 315 kB     00:00
```

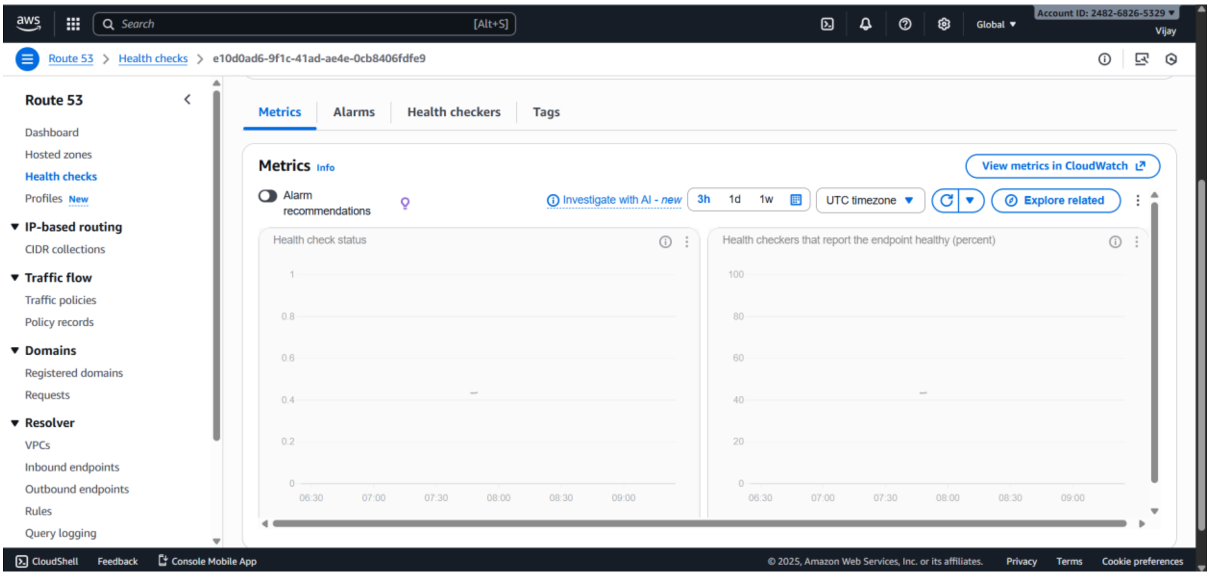
```
  Installing       : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                                         6/13
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                            7/13
  Running scriptlet: httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch                                    8/13
  Installing       : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch                                    8/13
  Installing       : httpd-core-2.4.65-1.amzn2023.0.2.x86_64                                          9/13
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                          10/13
  Installing       : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                                            11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                               12/13
  Installing       : httpd-2.4.65-1.amzn2023.0.2.x86_64                                              13/13
  Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64                                              13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                                  1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                             2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                                        3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                                     4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                5/13
  Verifying        : httpd-2.4.65-1.amzn2023.0.2.x86_64                                               6/13
  Verifying        : httpd-core-2.4.65-1.amzn2023.0.2.x86_64                                          7/13
  Verifying        : httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch                                    8/13
  Verifying        : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                                         9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                           10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                            11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                          12/13
  Verifying        : mod_lua-2.4.65-1.amzn2023.0.2.x86_64                                            13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                            apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          httpd-2.4.65-1.amzn2023.0.2.x86_64
  httpd-core-2.4.65-1.amzn2023.0.2.x86_64                    httpd-filesystem-2.4.65-1.amzn2023.0.2.noarch
  httpd-tools-2.4.65-1.amzn2023.0.2.x86_64                   libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                       mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl start httpd
[root@ip-172-31-72-108 ec2-user]# sudo systemctl enable httpd
```

```
[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
```


[root@ip-172-31-72-108 ec2-user]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Active: active (running) since Fri 2025-11-14 06:11:07 UTC; 21min ago
Docs: man:httpd.service(8)
Main PID: 25542 (httpd)
Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
Tasks: 177 (limit: 1053)
Memory: 13.3M
CPU: 1.166s
CGroup: /system.slice/httpd.service
├─25542 /usr/sbin/httpd -DFOREGROUND
├─25543 /usr/sbin/httpd -DFOREGROUND
├─25544 /usr/sbin/httpd -DFOREGROUND
├─25545 /usr/sbin/httpd -DFOREGROUND
└─25546 /usr/sbin/httpd -DFOREGROUND
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Nov 14 06:11:07 ip-172-31-72-108.ec2.internal httpd[25542]: Server configured, listening on: port 80
[root@ip-172-31-72-108 ec2-user]#


Get started Info
Choose your starting point
Register a domain
Transfer domain
Create hosted zones
Configure health checks
Configure traffic flow
Configure resolvers


Route 53 > Health checks > e10d0ad6-9f1c-41ad-ae4e-0cb8406fdfe9
Route 53
Dashboard
Hosted zones
Health checks
Profiles New
IP-based routing
CIDR collections
Traffic flow
Traffic policies
Policy records
Domains
Registered domains
Requests
Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Query logging
Metrics
Alarms
Health checkers
Tags
Metrics Info
View metrics in CloudWatch
Alarm recommendations
Investigate with AI - new
3h
1d
1w
UTC timezone
Explore related
Health check status
Health checkers that report the endpoint healthy (percent)
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

RESULT

Thus, successfully deployed and configured is a foundational web infrastructure: a dynamic EC2/Apache server is operational alongside a scalable, cost-optimized S3 static hosting platform, all centrally managed by AWS Route 53 which provides the domain name, intelligent routing policies, and proactive health checks to ensure high availability and automated failover capabilities while S3 Lifecycle Management optimizes storage costs.

EXP.NO: 04	Create and configure Amazon Elastic File System (EFS) to create scalable and shared file storage for AWS resources and EFS file system, mount it to EC2 instances, and configure access permissions
DATE: 13-11-25	

AIM

To successfully create, configure, and securely mount an Amazon Elastic File System (EFS) file system onto one or more Amazon EC2 (Linux) instances within the same Virtual Private Cloud (VPC), thereby establishing a highly available, scalable, and shared network file storage solution for distributed applications. The aim also includes setting up access control using Security Groups and, optionally, EFS Access Points and IAM policies.

PROCEDURE

1. Create EFS File System: Navigate to the EFS console, select "Create file system," choose your target VPC, and configure performance/storage settings.

2. Define Mount Targets: When creating the EFS, select the appropriate Availability Zones (AZs) to automatically create Mount Targets and attach a new or existing Security Group (e.g., EFS-SG) to these targets.

3. Configure EFS Security Group (EFS-SG): Edit the inbound rules of the EFS-SG to explicitly allow NFS traffic (Port 2049), specifying the Security Group of your EC2 instance(s) as the source.

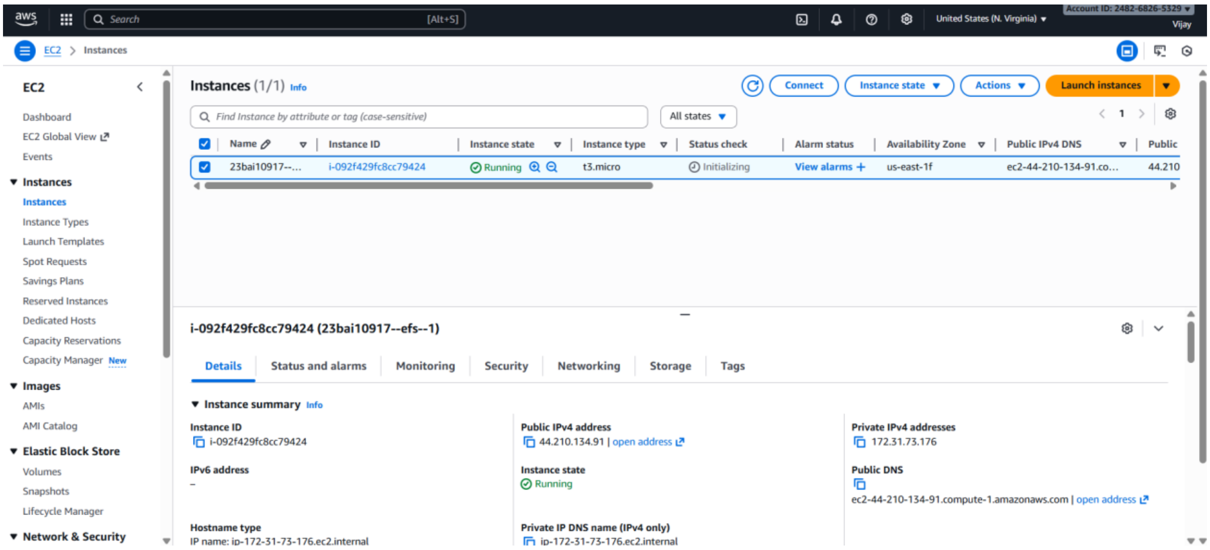
4. Verify EC2 Security Group (EC2-SG): Ensure the EC2-SG has an outbound rule that permits all traffic (or specifically Port 2049) destined for the EFS-SG to allow the connection initiation.

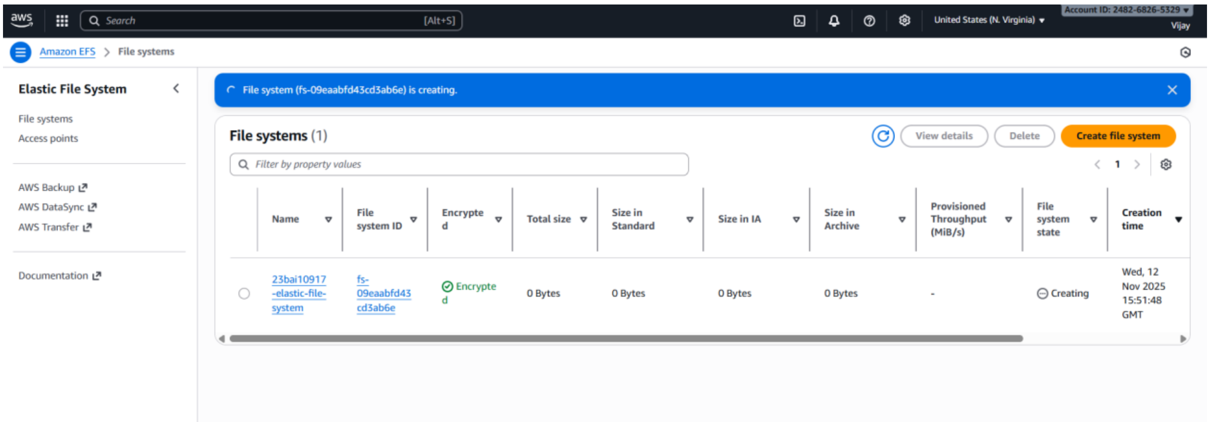
5. Connect and Install Utility: SSH into your EC2 instance and install the recommended Amazon EFS client utility (amazon-efs-utils) or the generic nfs-common package if using a non-Amazon Linux distribution.

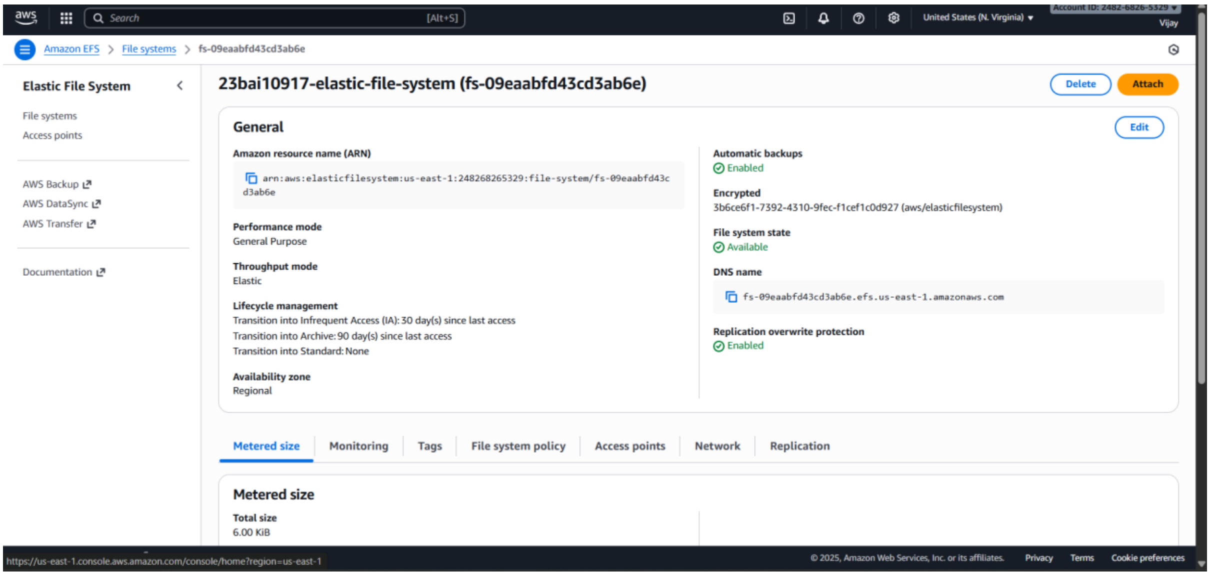
6. Create Mount Point: Create a local directory on the EC2 instance to serve as the mount point (e.g., /mnt/efs_share).

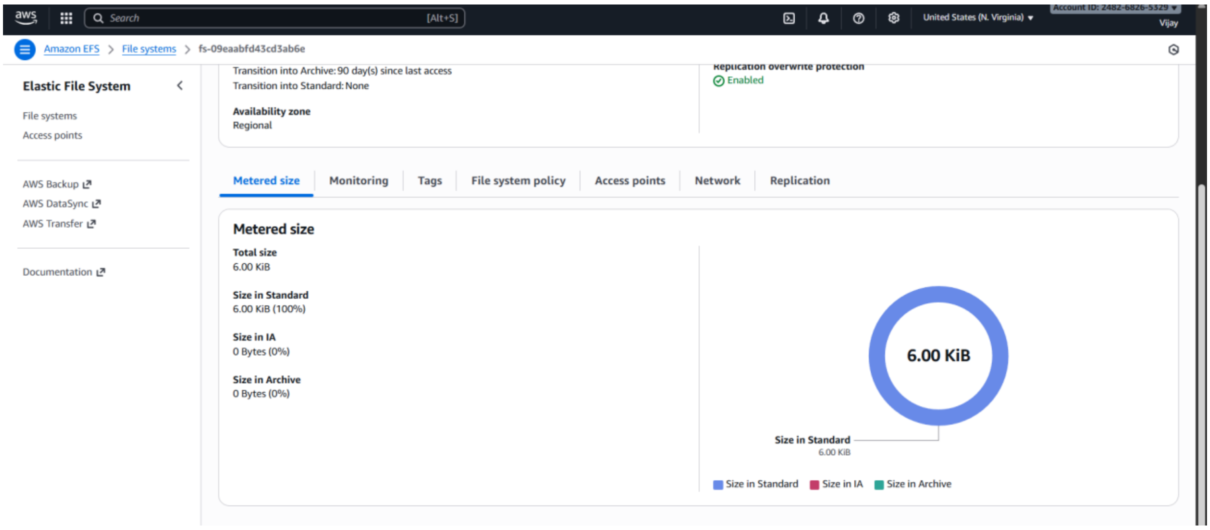
7. Mount the File System: Use the EFS Mount Helper command (found in the EFS console's "Attach" instructions) to execute the mount command: sudo mount -t efs -o tls fs-xxxxxxxx:/ /mnt/efs_share.

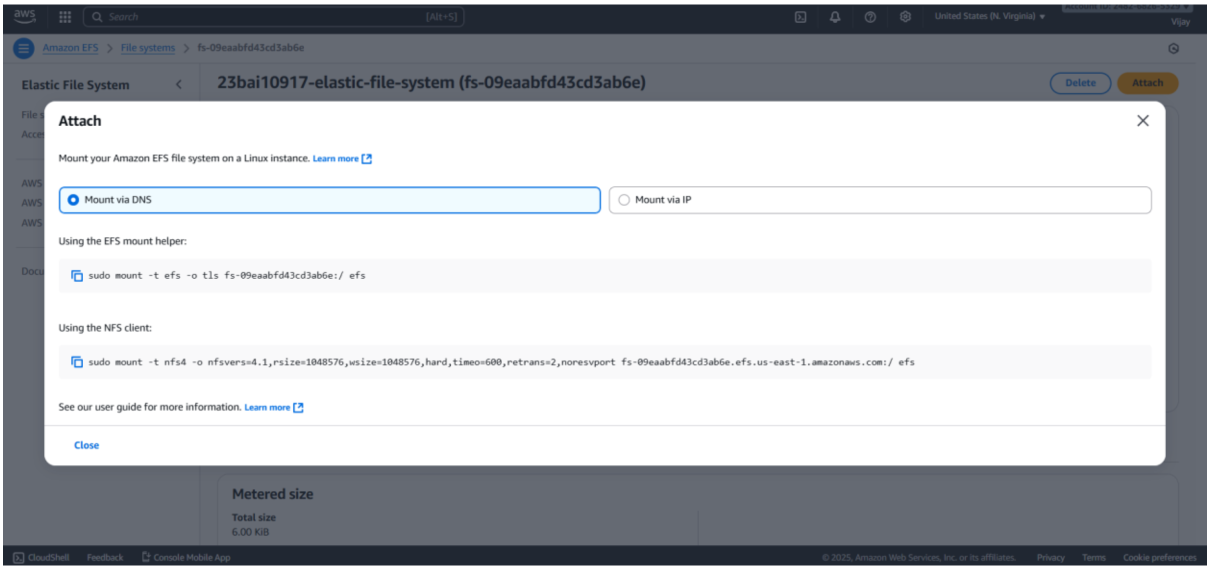
IMPLEMENTATION SCREENSHOT

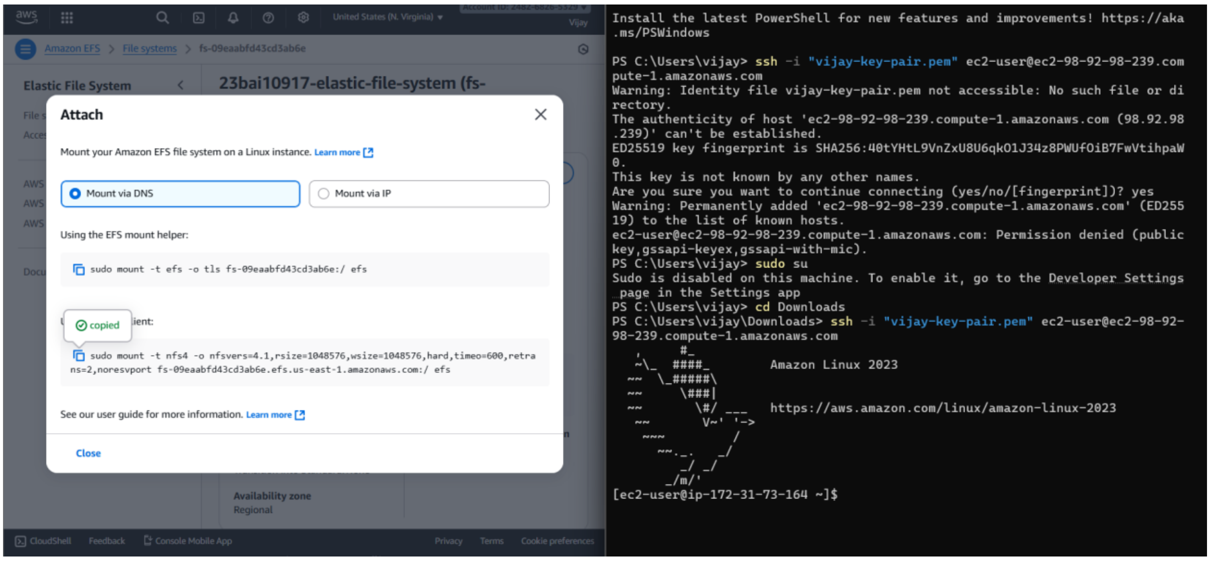





Search [Alt+S]
United States (N. Virginia)
Vijay
Amazon EFS > File systems > fs-09eaabfd43cd3ab6e
Elastic File System
File systems
Access points
AWS Backup
AWS DataSync
AWS Transfer
Documentation
23bai10917-elastic-file-system (fs-09eaabfd43cd3ab6e)
Delete
Attach
General
Edit
Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-1:248268265329:file-system/fs-09eaabfd43cd3ab6e
Performance mode
General Purpose
Throughput mode
Elastic
Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None
Availability zone
Regional
Automatic backups
Enabled
Encrypted
3b6ce6f1-7392-4310-9fec-f1cef1c0d927 (aws/elasticfilesystem)
File system state
Available
DNS name
fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com
Replication overwrite protection
Enabled
Metered size
Monitoring
Tags
File system policy
Access points
Network
Replication
Metered size
Total size
6.00 KiB
https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search [Alt+S]
United States (N. Virginia)
Vijay
Amazon EFS > File systems > fs-09eaabfd43cd3ab6e
Elastic File System
File systems
Access points
AWS Backup
AWS DataSync
AWS Transfer
Documentation
Transition into Archive: 90 day(s) since last access
Transition into Standard: None
Availability zone
Regional
Enabled
Metered size
Monitoring
Tags
File system policy
Access points
Network
Replication
Metered size
Total size
6.00 KiB
Size in Standard
6.00 KiB (100%)
Size in IA
0 Bytes (0%)
Size in Archive
0 Bytes (0%)
6.00 KiB
Size in Standard
6.00 KiB
Size in Standard
Size in IA
Size in Archive


Amazon EFS > File systems > fs-09eaabfd43cd3ab6e
Elastic File System
23bai10917-elastic-file-system (fs-09eaabfd43cd3ab6e)
Delete
Attach
Attach
Mount your Amazon EFS file system on a Linux instance. Learn more
Mount via DNS
Mount via IP
Using the EFS mount helper:
sudo mount -t efs -o tls fs-09eaabfd43cd3ab6e:/ efs
Using the NFS client:
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport fs-09eaabfd43cd3ab6e.efs.us-east-1.amazonaws.com:/ efs
See our user guide for more information. Learn more
Close
Metered size
Total size
6.00 KiB
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



RESULT

The EFS file system will be successfully mounted to the EC2 instance, verified by the presence of the EFS network volume when running the df -hT command. This result confirms that the storage is decoupled from the compute instance, allowing files written to the mounted directory to be immediately accessible, scalable, and persistent across all other EC2 instances securely mounted to the same EFS.

EXP.NO: 05	Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience
DATE: 13-11-25	

AIM

To Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience.

PROCEDURE

1. Select Distribution Type: Navigate to the CloudFront console and choose "Create Distribution," selecting the appropriate delivery method (usually Web for website content).

2. Define Origin: Specify the Origin Domain Name (e.g., your S3 bucket endpoint or Load Balancer DNS for EC2), set a unique Origin ID, and configure an Origin Access Control (OAC) to restrict direct public access to the Origin.

3. Configure Cache Behavior: Define the path patterns (e.g., Default (*)) and set the Viewer Protocol Policy (e.g., Redirect HTTP to HTTPS) and Allowed HTTP Methods (e.g., GET, HEAD).

4. Set Caching Policy: Choose or create a Cache Policy to control how long content stays in the edge locations (Time-To-Live or TTL) and which request headers, cookies, or query strings are forwarded to the Origin.

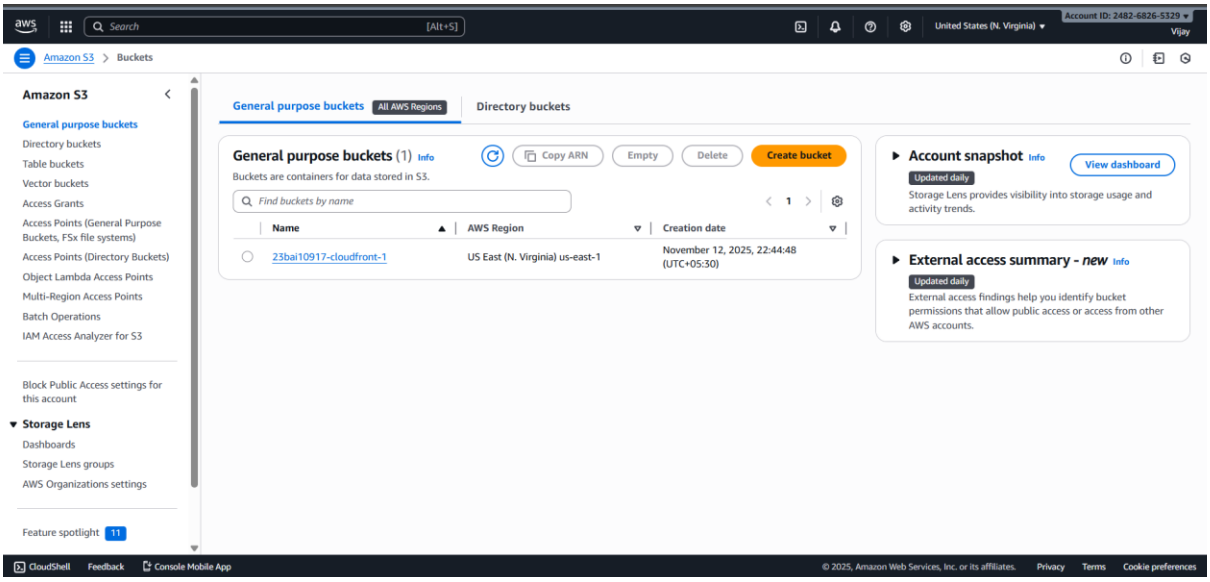
5. Distribution Settings: Specify the Price Class (geographic regions to serve), the Alternate Domain Names (CNAMEs) for your custom domain, and the Custom SSL Certificate for secure connections.

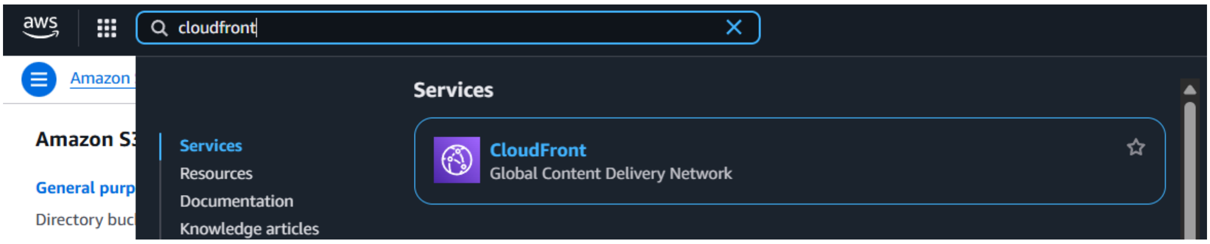
6. Create and Wait: Review all settings and click "Create Distribution." Wait for the distribution's status to change from InProgress to Deployed (this typically takes 15–30 minutes).

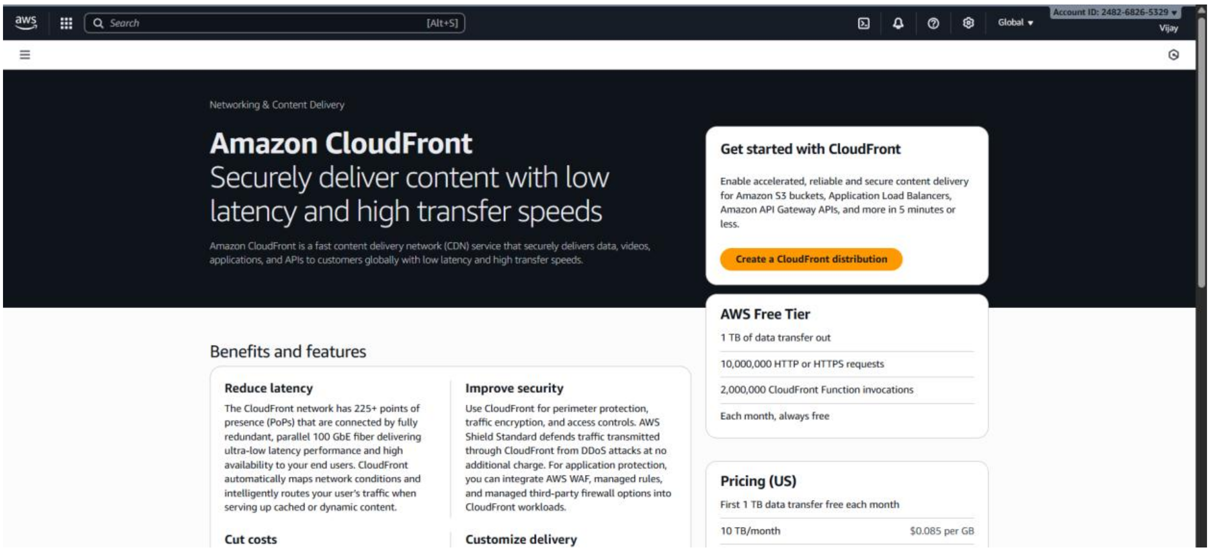
7. Update DNS: Once deployed, update your domain's DNS records by creating a CNAME record that points your custom domain name (e.g., www.example.com) to the new CloudFront Distribution Domain Name.

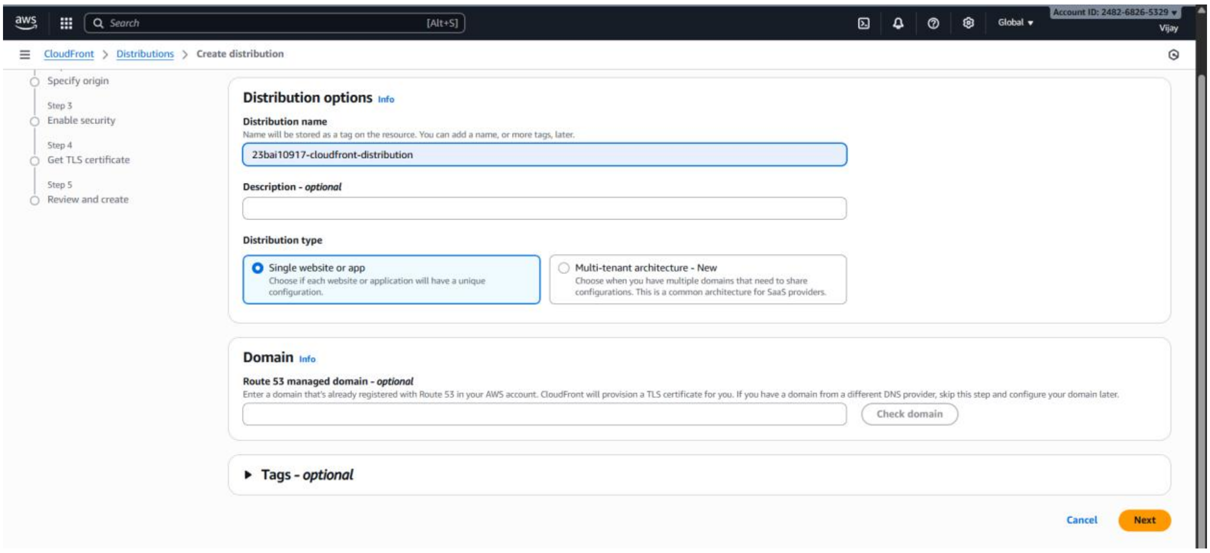
IMPLEMENTATION SCREENSHOT

Creating S3 bucket.

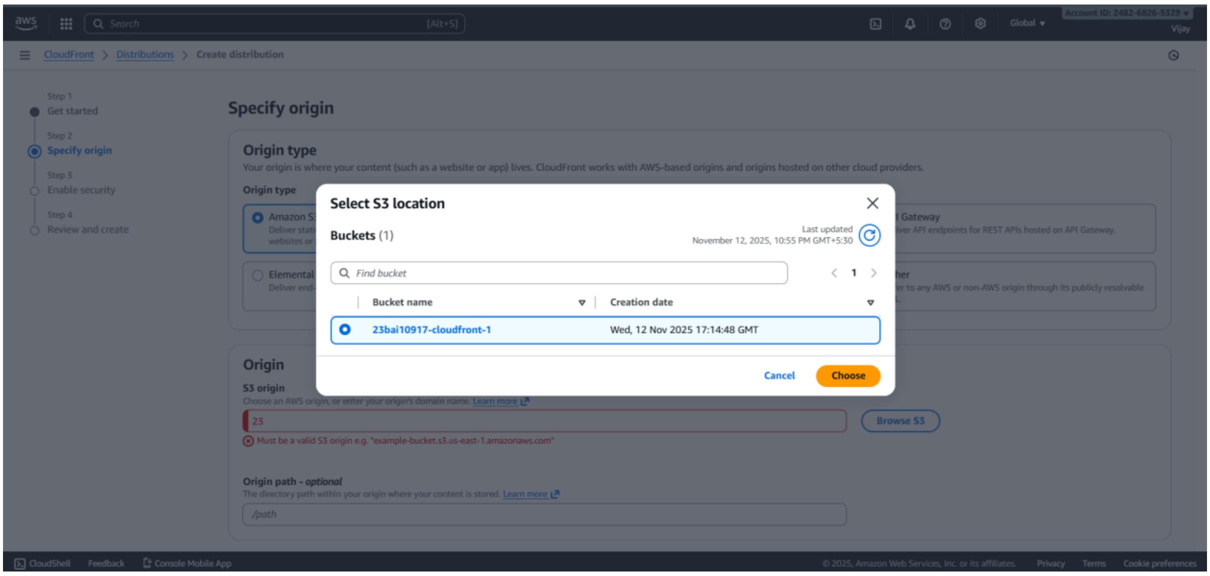






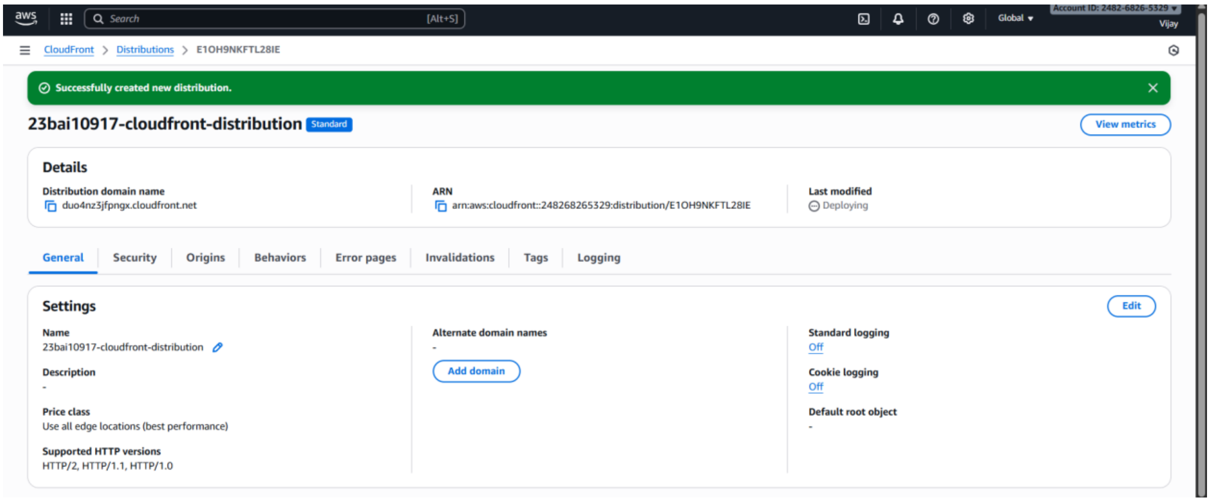


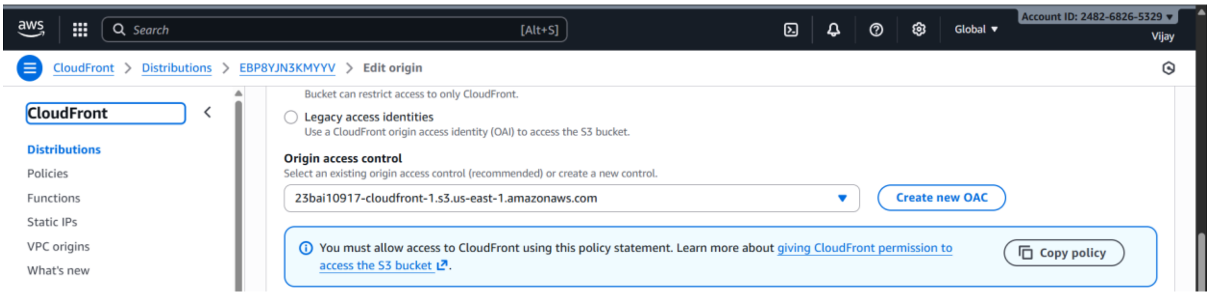
CloudFront > Distributions > Create distribution
Specify origin
Step 3
Enable security
Step 4
Get TLS certificate
Step 5
Review and create
Distribution options Info
Distribution name
Name will be stored as a tag on the resource. You can add a name, or more tags, later.
23bai10917-cloudfront-distribution
Description - optional
Distribution type
Single website or app
Choose if each website or application will have a unique configuration.
Multi-tenant architecture - New
Choose when you have multiple domains that need to share configurations. This is a common architecture for SaaS providers.
Domain Info
Route 53 managed domain - optional
Check domain
Tags - optional
Cancel
Next

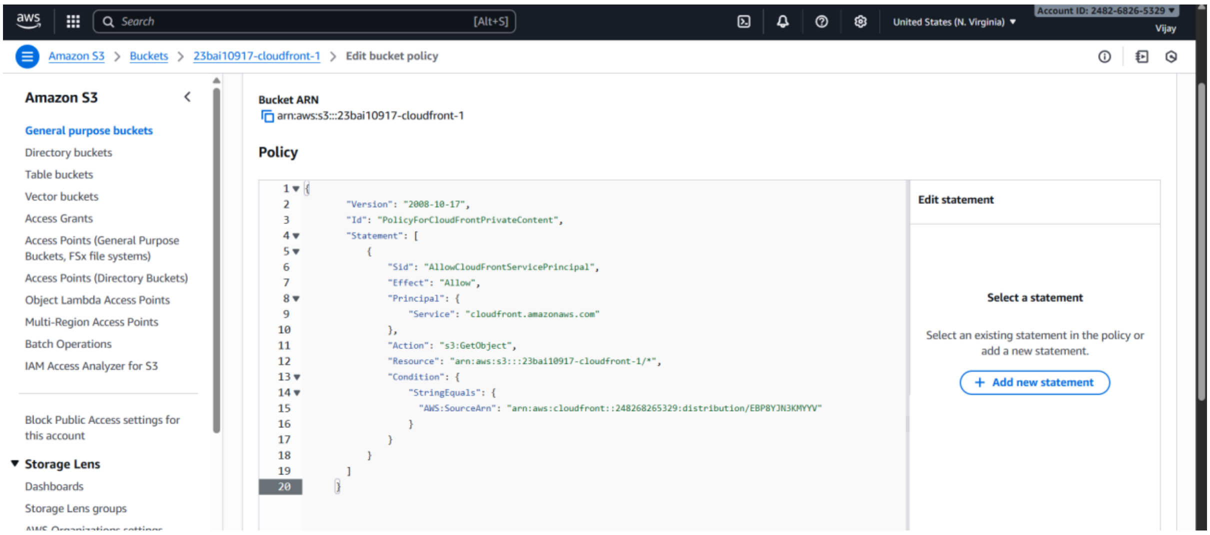


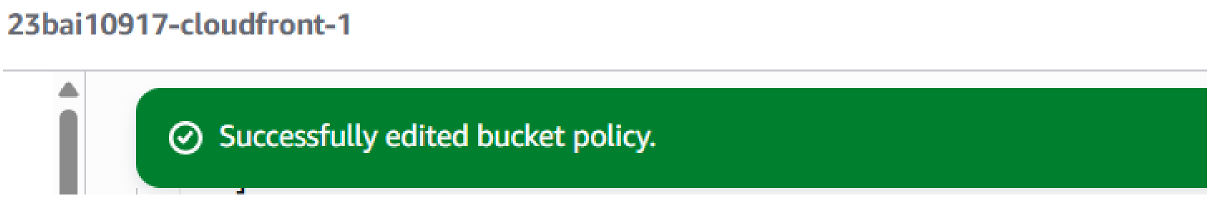
Specify origin
Origin type
Select S3 location
Buckets (1)
Last updated
November 12, 2025, 10:55 PM GMT+5:30
Find bucket
Bucket name
Creation date
23bai10917-cloudfront-1
Wed, 12 Nov 2025 17:14:48 GMT
Cancel
Choose
Origin
S3 origin
Browse S3
Must be a valid S3 origin e.g. "example-bucket.s3.us-east-1.amazonaws.com"
Origin path - optional
/path

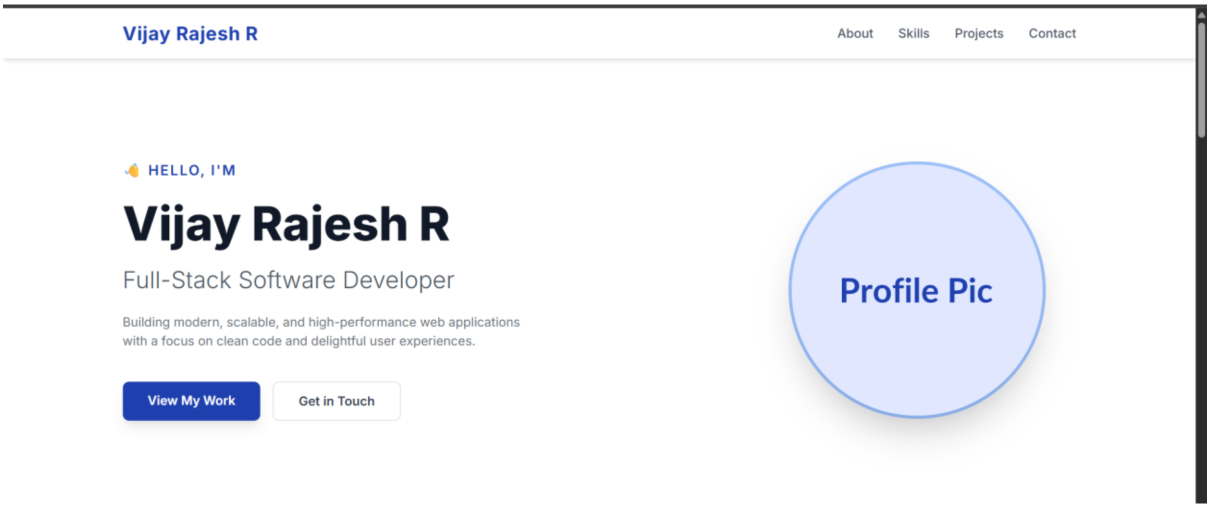
Origin
S3 origin
Choose an AWS origin, or enter your origin's domain name. Learn more
23bai10917-cloudfront-1.s3.us-east-1.amazonaws.com
Browse S3
Origin path - optional
The directory path within your origin where your content is stored. Learn more
/path


Search [Alt+S]
Global
Account ID: 2482-6826-5329
Vijay
CloudFront > Distributions > E1OH9NKFTL28IE
Successfully created new distribution.
23bai10917-cloudfront-distribution Standard
View metrics
Details
Distribution domain name
duo4nz3jfpngx.cloudfront.net
ARN
arn:aws:cloudfront::248268265329:distribution/E1OH9NKFTL28IE
Last modified
Deploying
General Security Origins Behaviors Error pages Invalidations Tags Logging
Settings
Edit
Name
23bai10917-cloudfront-distribution
Description
-
Price class
Use all edge locations (best performance)
Supported HTTP versions
HTTP/2, HTTP/1.1, HTTP/1.0
Alternate domain names
-
Add domain
Standard logging
Off
Cookie logging
Off
Default root object
-


Search [Alt+S]
Global
Account ID: 2482-6826-5329
Vijay
CloudFront > Distributions > EBP8YJN3KMYYV > Edit origin
CloudFront
Distributions
Policies
Functions
Static IPs
VPC origins
What's new
Bucket can restrict access to only CloudFront.
Legacy access identities
Use a CloudFront origin access identity (OAI) to access the S3 bucket.
Origin access control
Select an existing origin access control (recommended) or create a new control.
23bai10917-cloudfront-1.s3.us-east-1.amazonaws.com
Create new OAC
You must allow access to CloudFront using this policy statement. Learn more about giving CloudFront permission to access the S3 bucket.
Copy policy


Search [Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
Amazon S3 > Buckets > 23bai10917-cloudfront-1 > Edit bucket policy
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3
Block Public Access settings for this account
Storage Lens
Dashboards
Storage Lens groups
Bucket ARN
arn:aws:s3:::23bai10917-cloudfront-1
Policy
{
"Version": "2008-10-17",
"Id": "PolicyForCloudFrontPrivateContent",
"Statement": [
{
"Sid": "AllowCloudFrontServicePrincipal",
"Effect": "Allow",
"Principal": {
"Service": "cloudfront.amazonaws.com"
},
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::23bai10917-cloudfront-1/*",
"Condition": {
"StringEquals": {
"AWS:SourceArn": "arn:aws:cloudfront::248268265329:distribution/EBP8YJN3KMYYV"
}
}
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
Add new statement


23bai10917-cloudfront-1
Successfully edited bucket policy.



RESULT

Upon updating the DNS and successful propagation, website visitors accessing the custom domain will be automatically routed to the nearest CloudFront Edge Location, resulting in reduced latency, faster loading times, and a highly available content delivery experience. The distribution will effectively cache content, minimizing direct requests to the Origin and improving overall website performance and security.

EXP.NO: 06	Create AWS Certificate Manager (ACM) for managing SSL/TLS certificates for secure communication in AWS
DATE: 14-11-25	

AIM

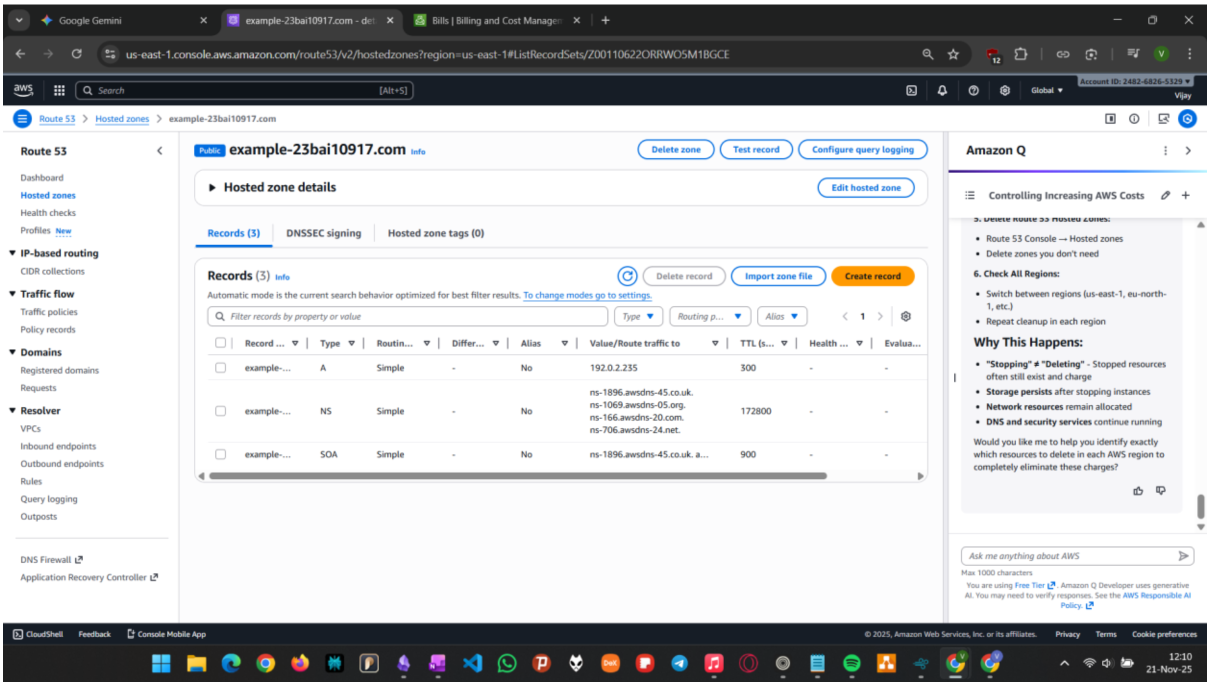
To **request a public SSL/TLS certificate** from AWS Certificate Manager (ACM), **validate domain ownership** using DNS records, and **integrate** the issued certificate with an **Application Load Balancer (ALB)** to enable secure **HTTPS communication** for a hosted application.

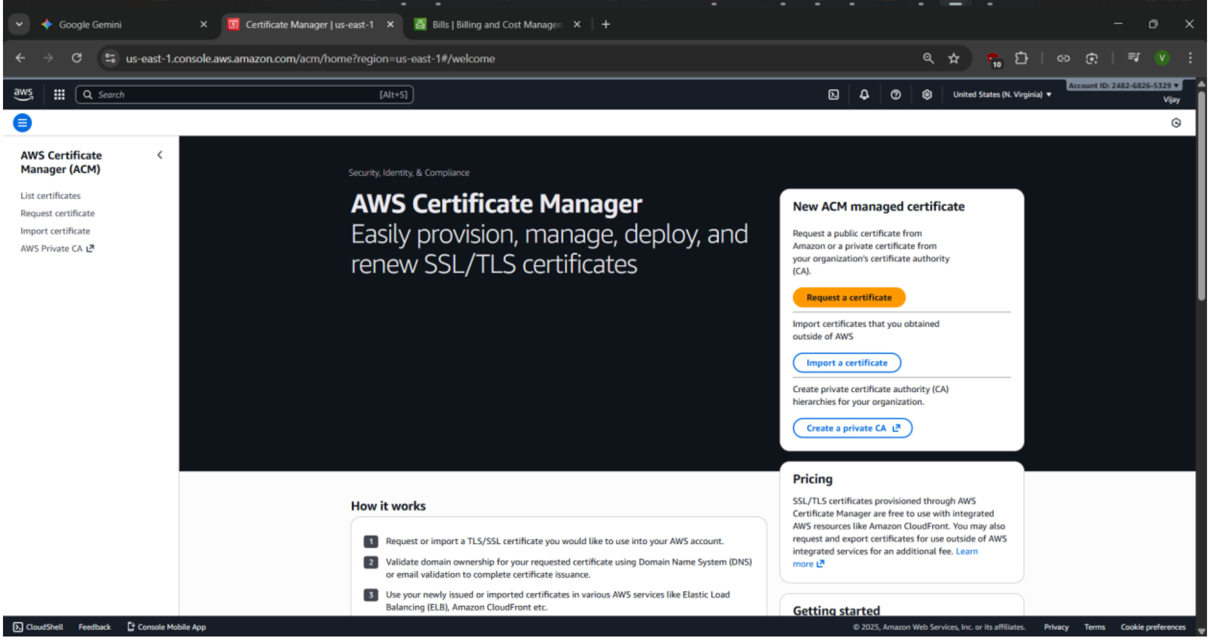
PROCEDURE

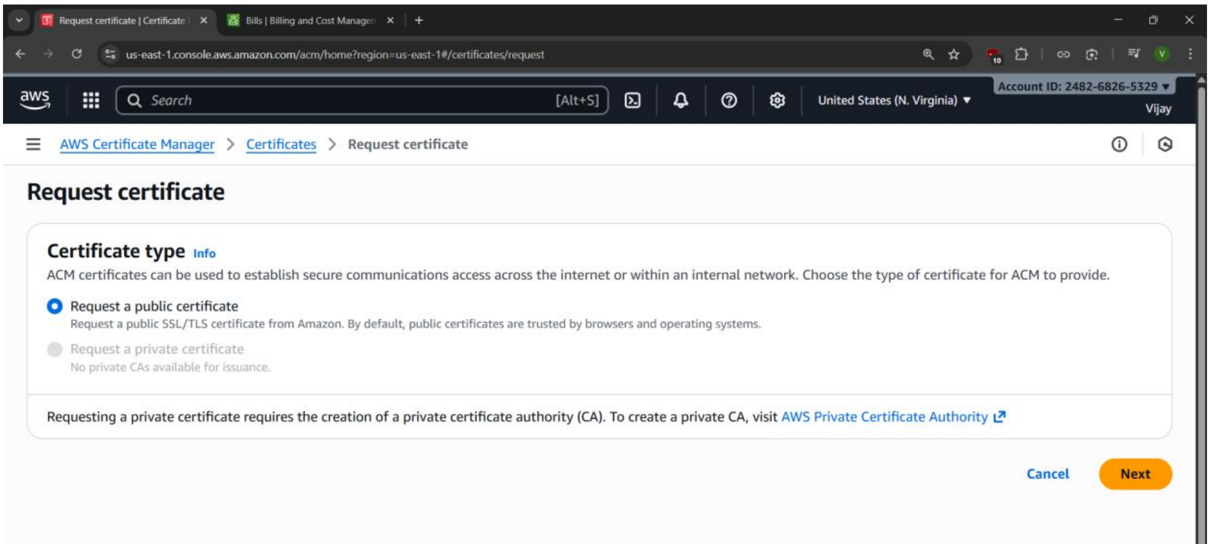
1. Request an ACM Certificate: Navigate to the ACM console and choose Request a public certificate. Enter the desired fully qualified domain name (FQDN) (e.g., www.example-lab.com) and a wildcard domain (e.g., *.example-lab.com) for multi-subdomain support. Select the DNS validation method and click Request.
2. Validate Domain Ownership (DNS): Once the certificate request is in Pending validation status, retrieve the CNAME record details provided by ACM. If the domain's DNS is hosted in Amazon Route 53, use the "Create records in Route 53" button to automatically add the required CNAME record to the public hosted zone. Wait for the certificate status to change to Issued.
3. Provision Target EC2 Instance: Launch a simple EC2 instance running a web server (like Apache or NGINX) to serve plain HTTP content on port 80. Create a Target Group in EC2 to register this instance as a target.
4. Create Application Load Balancer (ALB): Create an Application Load Balancer with listeners configured for both HTTP (port 80) and HTTPS (port 443). Configure the HTTP listener to redirect traffic to HTTPS.
5. Integrate ACM Certificate with ALB: For the HTTPS (port 443) listener on the ALB, under Default SSL/TLS certificate, select Choose a certificate from ACM and choose the Issued certificate from Step 2. Select a secure Security policy (e.g., ELBSecurityPolicy-2016-08).
6. Configure DNS Record for ALB: In Route 53, create an Alias record for the FQDN (www.example-lab.com) that points directly to the newly created ALB DNS name. This directs all public traffic for the domain to the Load Balancer.
7. Final Validation: Access the domain in a web browser using HTTPS (e.g., https://www.example-lab.com). Verify that the browser displays a secure connection (lock icon) and confirm the certificate details show the ACM-issued certificate. Attempting to access the

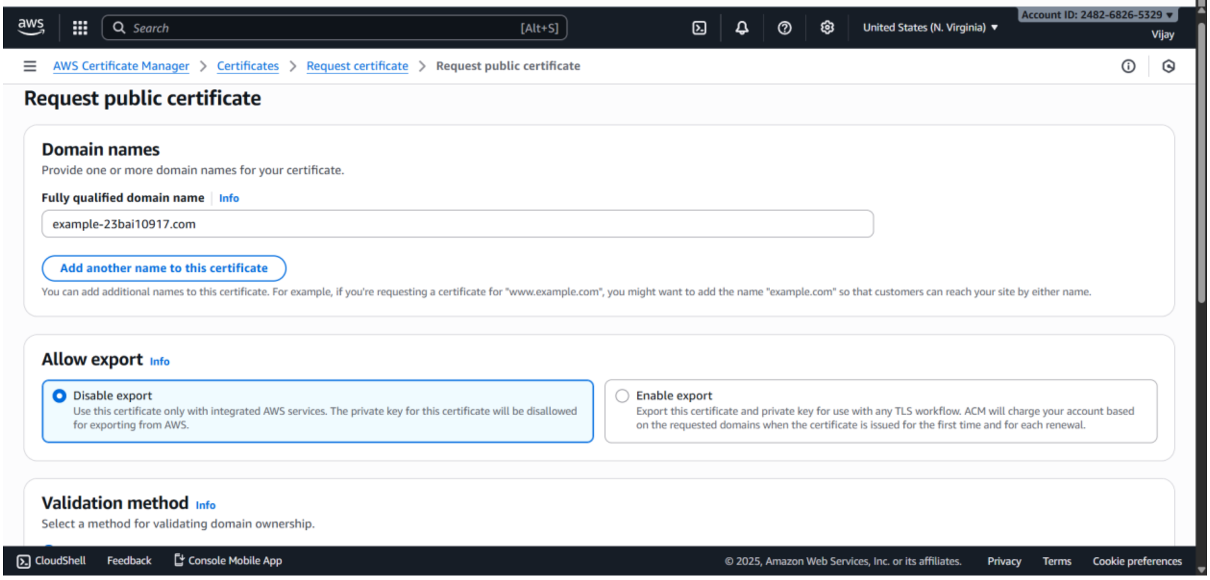
HTTP URL should automatically redirect to the secure HTTPS connection.

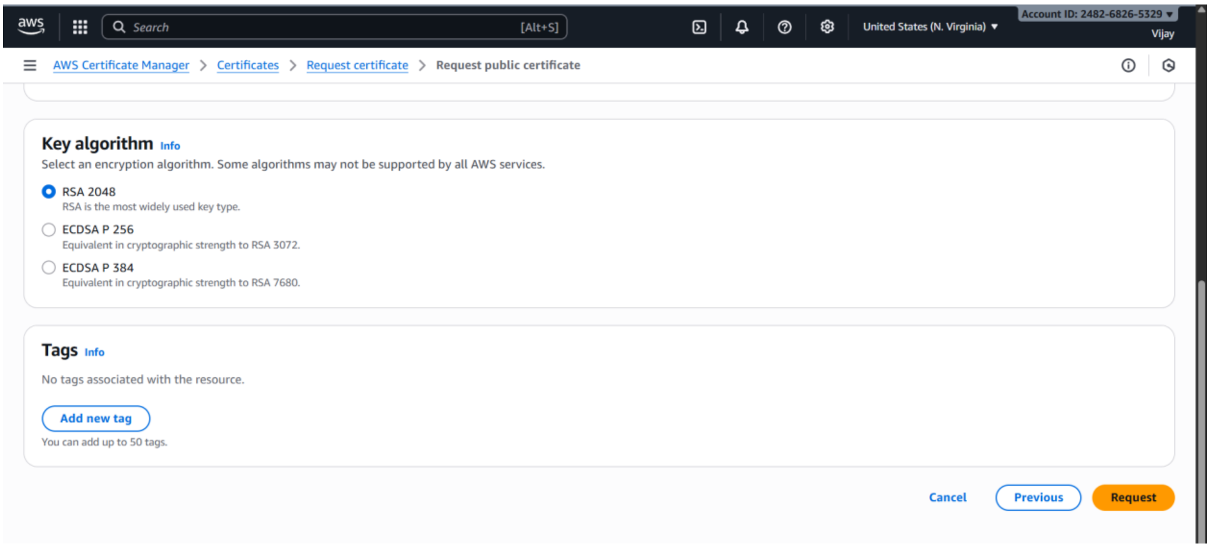
IMPLEMENTATION SCREENSHOT

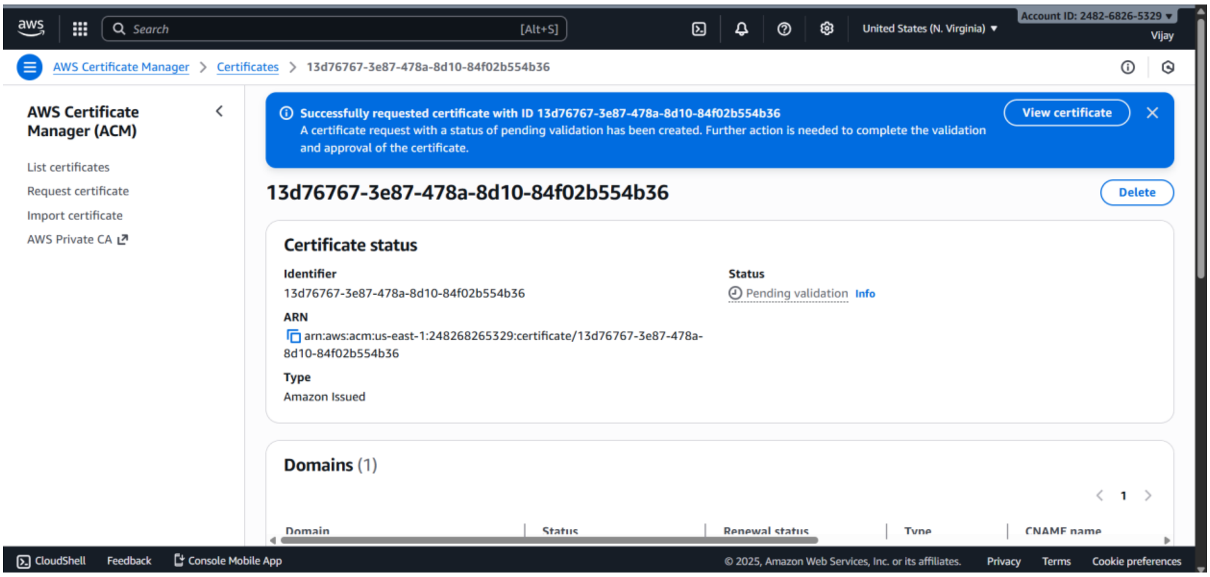





Request certificate | Certificate
Bills | Billing and Cost Managem
us-east-1.console.aws.amazon.com/acm/home?region=us-east-1#/certificates/request
Search
[Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
AWS Certificate Manager > Certificates > Request certificate
Request certificate
Certificate type Info
ACM certificates can be used to establish secure communications access across the internet or within an internal network. Choose the type of certificate for ACM to provide.
Request a public certificate
Request a public SSL/TLS certificate from Amazon. By default, public certificates are trusted by browsers and operating systems.
Request a private certificate
No private CAs available for issuance.
Requesting a private certificate requires the creation of a private certificate authority (CA). To create a private CA, visit AWS Private Certificate Authority
Cancel
Next


Search
[Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
AWS Certificate Manager > Certificates > Request certificate > Request public certificate
Request public certificate
Domain names
Provide one or more domain names for your certificate.
Fully qualified domain name Info
example-23bai10917.com
Add another name to this certificate
You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.
Allow export Info
Disable export
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.
Enable export
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.
Validation method Info
Select a method for validating domain ownership.
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
United States (N. Virginia)
Account ID: 2482-6826-5329
Vijay
AWS Certificate Manager > Certificates > Request certificate > Request public certificate
Key algorithm Info
Select an encryption algorithm. Some algorithms may not be supported by all AWS services.
RSA 2048
RSA is the most widely used key type.
ECDSA P 256
Equivalent in cryptographic strength to RSA 3072.
ECDSA P 384
Equivalent in cryptographic strength to RSA 7680.
Tags Info
No tags associated with the resource.
Add new tag
You can add up to 50 tags.
Cancel
Previous
Request



RESULT

The lab successfully demonstrated the automated provisioning of a publicly trusted SSL/TLS certificate via AWS Certificate Manager and its immediate integration with an Application Load Balancer. The certificate achieved the Issued status following DNS validation, and when integrated with the ALB, it enforced secure HTTPS termination for the web application, proving ACM's capability to manage secure communication with minimal manual intervention and ensuring managed certificate renewal.